“Sărac Lipit Pământului”: Land Restitution, Political Capture and the Fiscal Zombie in Rural Romania

Submitted in Partial Fulfillment of the Requirements for the Degree of
Master of Research in Economics

Sciences Po Paris, School of Research
Department of Economics

Titas Povilaitis

Under the Supervision of

Julia Cagé

Jury Member: **Ekaterina Zhuravskaya**

May 2026

"Sărac Lipit Pământului": Land Restitution, Political Capture and the Fiscal Zombie in Rural Romania*

By TITAS POVILAITIS†

May 2026

Abstract

This paper investigates the long-term economic and political consequences of Romania's 1991 land restitution and its interaction with the 2007 European Union accession shock. It argues that restitution-as-fragmentation helped to generate a mismatch between newly assigned smallholder property rights and the indivisible, large-scale socialist capital stock inherited from the collectivization era. Using Baseline and Synthetic Difference-in-Differences frameworks based on terrain ruggedness, the analysis demonstrates that this geographic asset mismatch was associated with a severe and persistent decline in mechanized capital – proxied by tractor density – across the historically fertile Plains. However, post-2007 economic recovery was determined by local institutions. While politically competitive plains successfully adapted through increasing rural enterprise, clientelist networks in captured counties were linked to considerably weaker commercial adaptation, with business formation being nearly half that of less-captured plains. Results are consistent with a "Fiscal Zombie" equilibrium, whereby gatekeeping of EU funds appears to have trapped an immobile population, with captured counties showing a 14% relative increase in fiscal dependency. In bridging spatial economics with institutional theory, this research shows that while geography bounds initial physical capacities, political capture may instrumentalize these vulnerabilities, producing patterns consistent with a captive, state-dependent electorate. (*JEL* P26, N54, R11, Q15, D72, H77)

*I am deeply grateful to my supervisor, Professor Julia Cagé, for her invaluable guidance and for enriching my academic journey throughout this thesis. I also want to thank Professor Ekaterina Zhuravskaya for her advice and for serving on my jury, and to Professor Paul Bouscasse for his instrumental help in shaping the initial concept of this research. My historical analysis of collectivization greatly benefited from the advice and archival expertise of Professors Gail Kligman and Katherine Verdery, as well as from Professor Dorin Dobrincu, whose feedback and suggested literature significantly deepened this project. I am also thankful to Valentina Marican at the National Institute of Statistics of Romania for her crucial assistance in locating essential agricultural data. Of course, I want to thank Prachi, Michelle, and my other friends for their unwavering support and encouraging discussions during this Master's program. Finally, my deepest gratitude goes to my parents – without them, I would not be where I am today. All errors are my own.

†titas.povilaitis@sciencespo.fr

Contents

I Introduction 3

II Related Literature 7

III Historical and Institutional Background 11
- III.A Pre-Collectivization Agrarian Landscape . . . 11
- III.B Collectivization, 1949-1962 . . . 12
- III.C Restitution & EU Accession . . . 16

IV Conceptual Framework 20

V Data 22

VI Identification Strategy 29
- VI.A Ordinary Least Squares . . . 29
- VI.B Baseline Difference-in-Differences & Dynamic Event Study . . . 30
- VI.C Split-Sample Difference-in-Differences . . . 31
- VI.D Synthetic Difference-in-Differences . . . 32

VII Results 33
- VII.A Geographic Baseline . . . 33
- VII.B Asset Mismatch Channel . . . 34
- VII.C Political Capture Channel . . . 37
- VII.D Quantifying the Fiscal Zombie . . . 38

VIII Robustness Checks 41
- VIII.A Extreme TRI Split . . . 41
- VIII.B Extreme PSD Split . . . 41
- VIII.C Truncated Timeline . . . 42
- VIII.D Leave-One-Out Analysis . . . 42

IX Conclusion 43

A Variable Construction and Sources 49

B Spatial Variation of Key Variables 51

C Robustness 53

D Identification Without Controls 60

I. Introduction

The collectivization of agriculture and its subsequent dismantling represent two of the most significant institutional shocks in the 20th century history of Eastern Europe. While historians have extensively documented the social and political dimensions of these transformations (Kligman & Verdery, 2011; Iordachi & Dobrincu, 2009), the ambiguous economic legacy they left behind – characterized by persistent inefficiency despite market reforms – has received comparatively less scrutiny in the economic literature. Romania, however, presents a distinct development paradox within this context, and one that is unusually tractable for empirical analysis. Unlike most transition economies, Romania's 1991 Land Law mostly returned to pre-collectivization owners in original strip parcels, without any consolidation mechanism, making the fragmentation of the inherited capital stock especially sharp and its geographic variation especially legible (Cartwright, 2017).

While the Southern and Eastern Plains contain some of the country's most fertile soil and historically served as an agricultural heartland, contemporary data reveal a visible underperformance relative to the rugged, less arable Central and Southwestern regions (Figure 1). Ioniță and Mărcuță (2023) document that the Northeast and Southeast regions, dominated by open plains, record the lowest average wheat production, whereas the fragmented, mountainous Center and Southwest record the highest. This inverts the geographic determinant of agricultural potential, suggesting that the country's historical granary operates persistently below its agronomic capacity. The underperformance extends beyond wheat – Plains counties record faster demographic decline through rural natural increase – the balance of births and deaths in rural areas –, and, among counties with deeper clientelist penetration, suppressed rural business formation – patterns that a purely agronomic account fails to explain.

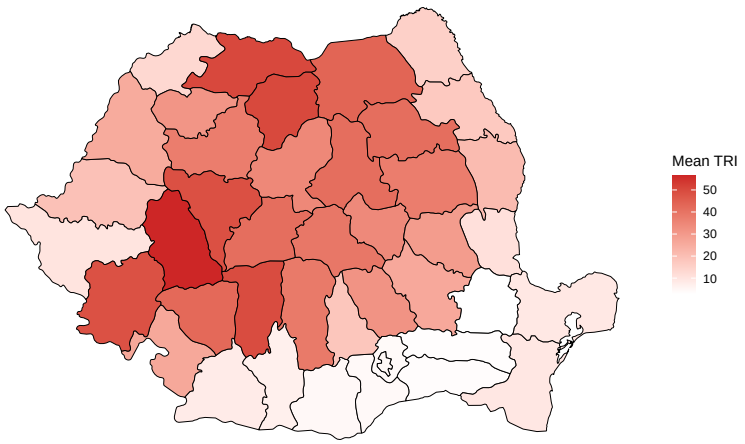


FIGURE 1: TERRAIN RUGGEDNESS INDEX ACROSS ROMANIAN COUNTIES

Notes: Larger Mean Terrain Ruggedness Index is associated with more rugged land.

This divergence challenges some standard transition models proposed by Swinnen and Rozelle (2006) as well as Gorton and Davidova (2004), which predicted that the restitution of private property rights would restore individual incentives and restructure agriculture toward market-oriented productivity. Instead, the Plains exhibit the reverse – The regions with the highest natural endowment for industrial-scale agriculture display not only lower productivity but also more pronounced demographic contraction than their mountainous counterparts[1], despite stronger private sector formation in the Plains[2] (Figure 2).

Furthermore, while Rizov et al. (2013) posit that Common Agricultural Policy (CAP) subsidies stabilized farm incomes after European Union (EU) accession in 2007, the persistence of this productivity and development gap implies that financial transfers may be masking structural inefficiencies rather than resolving them. The puzzle, then, is why the region whose geography and historical capital stock were most suited to large-scale mechanized agriculture became a locus of economic stagnation and low spatial mobility, while the naturally disadvantaged Mountain regions adapted comparatively more successfully to the market economy. This paper argues that the answer lies not in incomplete privatization, but in the specific interaction between 1991 restitution design and the physical capital stock that collectivization left behind.

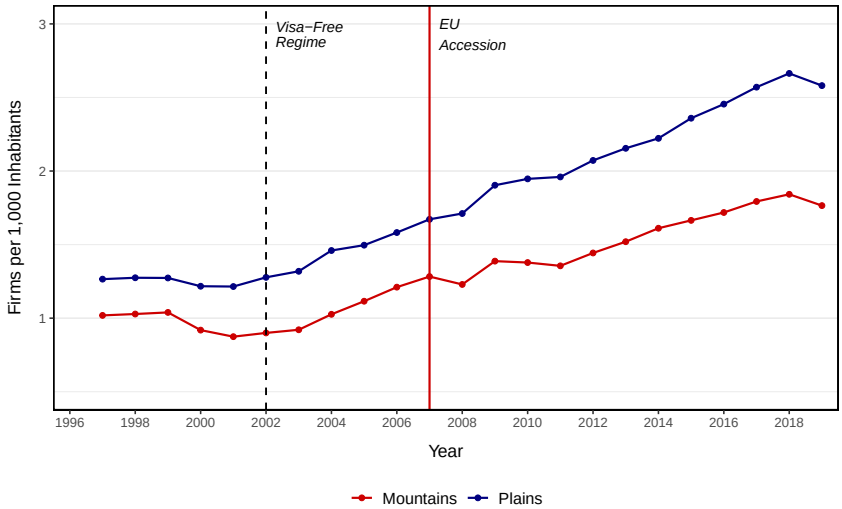


FIGURE 2: EVOLUTION OF RURAL BUSINESS FORMATION ACROSS GEOGRAPHY

Notes: Bucharest and the surrounding Ilfov county, which both belong to the Plains, are excluded from the sample.

The persistence of this regional disparity has crystallized into a stable political equilibrium rather than dissipating through market adjustment. The thesis refers to this configuration as a *Fiscal Zombie* state: a local administration that remains economically weak, structurally

[1]This dual dynamic inspired the thesis title, *"sărac lipit pământului"* ("poor, stuck to the earth")—a Romanian idiom serving as a double entendre for economic destitution and geographic immobility.

[2]Note that this aggregate effect is highly conditional on local political dynamics: Plains counties with high historical PSD support might actually exhibit suppressed rural business formation.

dependent on central fiscal transfers, and incapable of endogenous growth, yet retains political resilience precisely because that dependence creates a captive electorate. Existing political economy research establishes that low-productivity, high-inequality environments tend to favor clientelist forms of redistribution and public sector expansion as equilibrium political strategies, since these are more credible commitment devices than programmatic alternatives in contexts where enforcement of political promises is weak (Robinson & Verdier, 2013; Grzymala-Busse, 2008; O'Dwyer, 2006). This paper argues that the agricultural underperformance of the Plains constitutes the structural economic foundation of precisely this equilibrium.

Such a configuration has direct implications for democratic outcomes. The mechanism plausibly operates through spatial immobility: where exit from economically stagnant communities is constrained, voters lose the outside option that would otherwise discipline political actors (Hirschman, 1970). Without exit, populations become structurally dependent on state-mediated income sources, which creates the conditions under which access to social benefits is made politically conditional on electoral loyalty (Mares & Young, 2019; Stokes et al., 2013). The empirical implication, tested directly in this paper, is that Plains counties with deeper clientelist penetration – proxied by the vote share of the Social Democratic Party (PSD), the indirect successor of the Romanian Communist Party – should exhibit not only lower tractor density and, conditional on political capture, suppressed rural business formation, but also higher fiscal dependency on central transfers and accelerated demographic decline through out-migration and depressed rural natural increase. By linking the technological failure of decollectivization to these contemporary political and demographic outcomes, the paper shows how economic structure, rather than cultural predisposition, can sustain persistent patterns of political capture.

One of the key arguments of this thesis is that the failure of the Plains was not the result of incomplete privatization, but of a fundamental *asset mismatch* between restituted property rights and the inherited capital stock. The dissolution of socialist state farms dismantled a production system whose physical infrastructure – irrigation networks, storage facilities, and machinery – was highly specific to large-scale, vertically integrated agriculture. When land ownership was fragmented without a parallel mechanism for managing these indivisible assets, much of the existing capital effectively lost its productive value, collapsing to what Williamson (1987) terms its negligible next-best-use. Potentially productive infrastructure was either scrapped or captured by local elites, generating outcomes consistent with what De Soto (2001) describes as *dead capital*. Critically, this is distinct from the privatization failures documented elsewhere in the transition literature: Frydman et al. (1999) show that the method of privatization, not merely its occurrence, determines whether restructuring follows – and Romania's restitution-as-fragmentation design made productive reforms structurally unlikely from the outset.

The destruction of capital value created a second, reinforcing mechanism, which is a binding *liquidity constraint*. Although households in the Plains regained land ownership, the asset mismatch prevented them from converting land into liquid income. The land provided subsistence, but generated little marketable surplus, pushing households below the critical asset threshold below which productive investment and capital accumulation become self-reinforcing (Carter

& Barrett, 2006; Zimmerman & Carter, 2003). This is not a standard poverty trap – households were asset-rich by formal measures – but precisely the illiquid variant in which the type of asset, rather than its absence, binds the constraint.

This liquidity constraint translated into a condition of *forced immobility*. Migration, which requires upfront resources to cover relocation, job search, and initial settlement costs, remained inaccessible to households whose assets provided subsistence but little cash income – consistent with evidence that migration functions as an investment subject to financial constraints (Angelucci, 2015). Consequently, while transnational migration networks developed primarily in regions with lower asset specificity, particularly in and near the Mountains, where portable capital such as livestock and small machinery retained value after restitution (Sandu, 2005), the Plains population remained spatially locked. This asset-rich, cash-poor equilibrium produced the relatively stable and immobile demographic base that clientelist political arrangements require to persist. The geographic source of this variation – terrain ruggedness as a proxy for (de)collectivization intensity and therefore asset specificity – is exploited formally in the identification strategy.

This thesis makes three contributions. First, it contributes to the literature on institutional persistence by documenting a specific mechanism of technological lock-in that survives the formal transition to a market economy. While Acemoglu et al. (2001) and Nunn (2008) establish that historical shocks erode long-run development through cultural or legal channels, this paper argues that the *physical form of capital* can be an equally binding constraint. The asset mismatch created by restitution-as-fragmentation rendered the Plains' inherited infrastructure effectively worthless – an outcome that parallels Bleakley and Ferrie's (2014) finding that an exogenous land windfall generates no lasting productivity gain when the complementary capital needed to work that land is absent. In doing so, the paper challenges the standard transition view that privatization ensures restructuring (Swinnen & Rozelle, 2006; Frydman et al., 1999), showing instead that the design of property rights restitution can itself constitute a modernization trap.

Second, the paper contributes to the comparative literature on property rights and long-run agricultural performance. Markevich and Zhuravskaya (2018) show that where land following the abolition of Russian serfdom was assigned to repartition communes rather than directly to individual cultivators, agricultural productivity remained depressed for decades – partly because the governance structure of property rights, not merely their formal existence, determines investment incentives. The parallel is precise: Romania's 1991 restitution returned formal ownership to households while leaving the governance of indivisible collective infrastructure unresolved, producing an analogous suppression of productive capacity. This thesis shows that the technological legacy of 1950s collectivization continues to shape political behavior in the 2000s, offering a structural rather than cultural explanation for the Fiscal Zombie equilibrium – and thereby extending Pop-Eleches' (2007) account of post-communist path dependence from institutional to physical capital.

Third, the paper contributes methodologically by deploying a sequential identification strategy that isolates the asset mismatch mechanism from confounding national-level trends. Begin-

ning with Ordinary Least Squares (OLS) specification that establishes the geographic baseline, the paper proceeds through event studies validating parallel pre-trends in most cases, pooled and split-sample difference-in-differences (DiD) designs that distinguish the asset channel from the political capture channel, and concludes with a Synthetic Difference-in-Differences (SDiD) estimator (Arkhangelsky et al., 2021) applied to the high-capture subsample. This final estimator is the only specification that recovers a statistically significant fiscal dependency effect, which the paper interprets as evidence that SDiD's unit-weighting successfully filters national structural noise to isolate genuine political capture variation. Terrain ruggedness – a geographic feature that determined (de)collectivization intensity but is otherwise plausibly orthogonal to post-1991 outcomes – serves throughout as the identification device.

The remainder of this paper is organized as follows: Section II reviews the related literature on transition economics, asset specificity, and clientelism. Section III provides the historical and institutional background of Romanian (de)collectivization. Section IV explains the conceptual framework, Section V outlines the data and Section VI introduces the identification strategy. Section VII and VIII present the results and robustness checks, respectively, and Section IX concludes.

II. Related Literature

The post-socialist transition in agriculture was significantly predicated on the assumption that dismantling collective structures would unleash individual incentives and restore efficiency. However, the empirical record on the superiority of small-scale family farming in transition economies remains ambiguous. While some early studies suggested that individual farms could outperform corporate successors under specific conditions (Gorton & Davidova, 2004), more recent evidence indicates that external financial supports, such as CAP subsidies, discussed in the Romanian context in Section III, may be artificially sustaining productivity figures rather than reflecting genuine structural improvements (Rizov et al., 2013). Critically, this ambiguity is not merely about farm organization: Frydman et al. (1999) show that the method of privatization determines whether restructuring follows, suggesting that the design of restitution, not just its occurrence, is the relevant variable.

An important proximate explanation for post-restitution underperformance is the fragmentation of land ownership. The macro-development literature identifies a well-documented causal link between farm size and agricultural productivity: reducing farm size below a critical threshold can produce productivity losses of up to 17% through the misallocation of resources and the distortion of technology adoption decisions (Adamopoulos & Restuccia, 2020). This scale penalty is particularly acute in mechanized agriculture, where the proliferation of small holdings creates a landscape of farms that is individually too small to exploit the capacity economies of modern machinery (Foster & Rosenzweig, 2022). In Eastern Europe, this fragmentation has been identified as a barrier to rural development that hampers not only production, but also the functioning of land markets themselves (Hartvigsen, 2014; Sabates-Wheeler, 2002).

However, the fragmentation alone cannot explain the asymmetric outcomes documented in

this paper, since land restitution in Romania fragmented holdings in both the Plains and the Mountains. The critical difference lies in the nature of the capital stock that restitution inherited. In the Mountains, the pre-collectivization agricultural system had relied predominantly on low-specificity, portable assets – livestock, small-scale equipment, orchard cultivation – that retained productive value at the household scale after fragmentation. In the Plains, by contrast, the collectivized system had been built around highly specific, indivisible infrastructure – large-scale irrigation networks, mechanized grain processing, centralized storage – whose productive value was conditional on the scale and vertical integration of the socialist farm. Mathijs and Swinnen (1998), comparing decollectivization outcomes across Central and Eastern Europe, find that regions with less capital-intensive and more mixed agricultural systems adapted to private farming faster and with fewer productivity losses – a pattern directly consistent with the Plains-Mountains divergence documented here.

This failure of land reform to automatically generate growth is consistent with comparative evidence from other post-socialist contexts. In Russia, the dissolution of state farms produced considerable declines in grain production and capacity utilization rather than a market-led boom (Wegren, 2005). Reassessments of the Chinese experiment suggest that the Household Responsibility System was not the sole driver of agricultural growth, with price liberalization and pre-existing trajectories accounting for much of the observed improvement (Bramall, 2004; Ferguson & Kim, 2023). In the Romanian context, Verdery (2003) argues that restitution treated land as a political symbol and a complex of social relations rather than a productive asset, generating what she terms "fuzzy property" rights that discouraged consolidation and investment. This political logic of restitution – returning land to original strip parcels without any mechanism for managing inherited collective infrastructure – is precisely what the asset mismatch mechanism formalized in this paper seeks to explain.

Fragmentation, then, explains the inefficiencies related to scale, but does not fully capture the destruction of value that occurs when the physical capital stock is structurally incompatible with the new property rights regime. Transaction cost economics provides the relevant framework. Williamson (1987) identifies asset specificity – the degree to which a durable investment loses value when redeployed outside its original governance structure – as the critical determinant of economic organization: when the specific transaction or organizational structure supporting an asset dissolves, its value collapses to a negligible next-best use. Klein et al. (1978) as well as Grossman and Hart (1986) demonstrate that when assets are strictly complementary – as large-scale irrigation infrastructure and monoculture grain production are – vertical integration is the efficiency-maximizing arrangement precisely because it prevents the opportunistic behavior that specificity invites. When that integration is abruptly dissolved by restitution, the complementary assets become effectively worthless to the fragmented private owners who inherit them. Gow and Swinnen (1998), analyzing the Romanian and broader transition context directly, identify this as a hold-up problem: new private landowners possessed the soil, but processing and service infrastructure was privatized separately or abandoned, breaking the supply chain and eliminating the investment incentives that vertical integration had previously secured.

The hold-up problem has a direct technological consequence. Manuelli and Seshadri (2014)

argue that the diffusion of modern agricultural inputs – tractors being the canonical case – is not automatic but contingent on the relative price of labor: only when wages rise sufficiently does it become unprofitable to maintain labor-intensive alternatives. When the hold-up problem simultaneously devalues existing machinery and renders new capital investment risky through insecure property rights (Besley, 1995; Deininger & Jin, 2006), producers may rationally regress toward labor-intensive techniques even as the broader economy modernizes. In the Plains context, this mechanism predicts a persistent tractor deficit relative to what terrain and agronomic potential would otherwise imply – and it is precisely this deficit that the empirical analysis documents.

The long-term consequence of this capital destruction is the formation of asset-based poverty traps. Carter and Barrett (2006) as well as Zimmerman and Carter (2003) theorize the existence of a critical asset threshold below which households are unable to generate the surplus necessary for capital accumulation, becoming structurally locked in low-level equilibrium through a process of bifurcated accumulation dynamics: households above the threshold invest and grow, while those below dissave toward subsistence steady state. Crucially, this trap is not standard income poverty. What distinguishes the Romanian Plains case is the type of asset involved. Hornbeck and Naidu (2014) show that the destruction of capital – in their case through the 1927 Mississippi flood – drives out-migration as households seek returns elsewhere. The shock analyzed in this paper operates through a reverse mechanism: restitution granted land, a formally valuable asset, while simultaneously destroying the complementary capital that made that land productively liquid. The result is an asset-rich, cash-poor equilibrium in which the poverty trap binds labor to the land rather than releasing it. This distinction – immobility as the consequence of an illiquid asset endowment rather than asset absence – is central to the paper's argument and differentiates it from standard accounts of post-socialist rural decline.

This paper situates these mechanisms within the broader literature on institutional persistence and long-run development, while making a specific claim about the channel through which historical shocks bind. Acemoglu et al. (2001) and Nunn (2008) establish that critical junctures – whether colonization strategies or the slave trade – create institutional legacies that persist for centuries through cultural and legal channels. In the context of agriculture specifically, Banerjee and Iyer (2005) show that colonial land tenure systems in India continue to explain modern gaps in health and education investment, and Dell (2010) demonstrates that the Peruvian mita depresses consumption and public goods provision in affected districts to present day. These papers establish the general plausibility of persistence, but the channel in each case is institutional – property rights regimes, legal structures, or governance arrangements. This paper argues that *physical capital* can be an equally binding persistence channel. The most precise comparator is Bleakley and Ferrie (2014), who show that an exogenous land windfall on the Georgia frontier produced no lasting productivity gain – because land without the complementary capital and market infrastructure needed to work it at scale generates no durable accumulation dynamic. The Romanian restitution shock is structurally similar: formal property rights were restored, but the physical infrastructure that gave those rights productive value was simultaneously destroyed.

In Eastern Europe, persistence mechanisms often operate along the lines of former imperial

borders, a pattern directly relevant to the Romanian case. Grosfeld and Zhuravskaya (2015) and Pop-Eleches (2007) provide evidence that pre-industrial partitions between the Russian, Austrian, and Prussian empires continue to predict voting behavior and infrastructure density today. In Romania specifically, the Carpathian Mountains served as the approximate boundary between Habsburg-administered Transylvania and the Ottoman-influenced Principalities, producing pre-existing institutional divergences this paper analyzes. Heldring et al. (2015) demonstrate that structural shocks to property rights – in their case the dissolution of English monasteries – can catalyze lasting reorderings of labor markets and commercialization patterns. Taken together, this body of work suggests that the shock of decollectivization should be understood not as a policy reset on a blank slate, but as a critical juncture whose effects are conditioned by, and interact with, the institutional legacies of the collectivization era itself – a point that motivates the historical background developed in Section III.

The final link in the causal chain concerns demographic and political consequences of the asset-rich, cash-poor equilibrium. Standard spatial equilibrium models predict that labor should flow from low- to high-productivity areas until returns are equalized, but the evidence points to considerable friction in this adjustment. Bryan et al. (2014) show experimentally that households systematically under-invest in migration not due to a lack of potential returns but due to risk aversion and the inability to cover upfront costs – a liquidity constraint that produces involuntary immobility. Dustmann and Görlach (2016) further note that temporary migration decisions require liquid assets that subsistence agriculture characteristically fails to generate. In the Romanian context, Sandu (2005) shows that migration is heavily network-dependent: communities without early transnational pioneers face compounding barriers to exit, reinforcing regional disparities once they emerge. Dutronc-Postel and Zhuang (2024) add that perceived land insecurity can itself depress economic performance by keeping households oriented toward asset defense rather than productive investment or mobility – a dynamic consistent with the insecure property environment that fuzzy restitution created in the Plains.

When populations remain immobile and economically vulnerable, they generate specific incentives for political actors. The foundational logic is Hirschman's (1970): where exit is unavailable, populations can exercise only voice – and where voice is itself captured by incumbent networks, the result is political lock-in rather than accountability. Iyer and Mani (2012) provide empirical grounding for this mechanism, showing that bureaucrats and politicians exercise greater extractive control over immobile citizens precisely because the threat of exit cannot discipline them. This facilitates the form of state capture described by Grzymala-Busse (2008), in which political parties "colonize" public administration to extract rents, manifesting as the runaway state-building that O'Dwyer (2006) documents – the expansion of public sector employment as patronage rather than service delivery. Stokes et al. (2013) as well as Mares and Young (2019) provide the micro-foundations of this dynamic: brokers and local elites target immobile, welfare-dependent voters with conditional transfers, making benefit access contingent on electoral loyalty, and thereby locking dependent populations into a stable clientelist equilibrium.

Taken together, this literature traces a coherent causal chain that the paper tests empirically. The asset mismatch created by restitution-as-fragmentation destroyed the productive value

of Plains infrastructure, pushing households below the asset threshold for capital accumulation. The resulting liquidity constraint prevented migration, producing the spatially stable and economically dependent population that clientelist networks require. Political capture then transformed this economic stagnation into a durable political equilibrium – the Fiscal Zombie state – in which fiscal dependence on central transfers and concentrated PSD vote share reinforce each other. The empirical strategy developed in Section VI is designed to isolate each link in this chain, using terrain ruggedness as a geographic proxy for (de)collectivization intensity and the 2007 EU accession shock as an exogenous intensifier that allows dynamic identification of the divergence between Plains and Mountain counties.

III. Historical and Institutional Background

A. Pre-Collectivization Agrarian Landscape

Romania's agrarian landscape on the eve of collectivization was already defined by geographic duality that the socialist transformation would later exploit, entrench, and ultimately bequeath to the post-communist period. The interwar land reforms of 1921, which redistributed approximately 6 million hectares from large estates to peasant smallholders, had produced a formally egalitarian but structurally fragmented agricultural sector.[3] Yet this fragmentation manifested differently across terrain. In the Southern and Eastern Plains – the historical Bessarabia and Old Kingdom lowlands – pre-reform estates had been larger, relatively more mechanized, and oriented toward grain monoculture for export, meaning that post-reform fragmentation produced holdings that were individually too small to sustain the capital-intensive farming the terrain demanded. Gusti's (1938) *Enciclopedia României* records average smallholder (*improprietăriți*) holdings as of 1934 at approximately three hectares in the Plains regions (Old Kingdom and Bessarabia) and just one hectare in the Mountains (Transylvania and Bucovina at the time) – a distribution that already signals a limitation of holding size alone as an explanatory variable for post-transition outcomes, since Mountain smallholders operated on smaller plots yet adapted more successfully after 1991 (Gusti, 1938).[4] At the other end of the distribution, large landowners averaged approximately 400 hectares in the Plains against 140 hectares in the Mountains – a scale differential that had already generated an industrial logic in Plains agriculture, with large estates requiring mechanized equipment, irrigation infrastructure, and vertically integrated processing arrangements that smallholder Mountain farming had developed relatively less (Foster & Rosenzweig, 2022; Adamopoulos & Restuccia, 2020).

In the Mountains, by contrast, the pre-collectivization system was built around mixed, low-capital agriculture – livestock rearing, orchard cultivation, small-scale dairy – whose productive logic was household-scale and whose capital stock was correspondingly portable and

[3]The 1921 reform was the largest land redistribution in interwar Eastern Europe by area. See Henry L. Roberts, *Rumania: Political Problems of an Agrarian State* (New Haven: Yale University Press, 1951), pp. 12–18.

[4]The 1934 data on landholding structure by region remains one of the few pre-communist sources with region-level granularity. Direct digitization of equivalent census-level agricultural holdings tables was confirmed as unavailable by Professors Kligman and Verdery in correspondence with the authors (November 2025), who noted that raw figures would require archival recovery in Romania.

divisible. A family's livestock, its fruit trees, its small plot of maize did not require vertical integration with processing infrastructure to retain productive value. In the Plains, however, even smallholder agriculture operated in proximity to, and partial dependence on, the large-estate infrastructure that dominated the landscape. This asymmetry in the nature of the pre-existing capital stock – scale-specific and infrastructurally dependent in the Plains, portable and self-contained in the Mountains – is the baseline condition from which all subsequent divergence flows. Collectivization did not create this duality, but it built an entirely new and more rigid layer of fixed, specific capital on top of it in the Plains, and the manner of its dismantling after 1989 would determine which regions could reconstitute productive agriculture and which could not. This and other significant institutional transformations, starting from the 1949 collectivization decree through the 1991 restitution legislation and the 2007 EU accession shock will not only be discussed in this section, but will also be situated and summarized chronologically in Figure 3.

B. Collectivization, 1949-1962

Romanian agricultural collectivization was initiated by decree in March 1949, following the consolidation of Communist Party control over the state apparatus and the formal adoption of the Soviet model of forced collectivization as official policy.[5] The campaign proceeded in three broadly identifiable phases: an initial coercive push between 1949 and 1953, a tactical retrenchment following Stalin's death and the 1953 New Course, and a final accelerated drive between 1957 and 1962 that achieved formal completion.[6] By June 1962, the Romanian state announced that 94% of arable land had been incorporated into collective agricultural units (*Cooperative Agricole de Producție*, CAP) or state farms (*Întreprinderi Agricole de Stat*, IAS), making Romania's collectivization one of the most complete in the Eastern Block outside the Soviet Union itself (Kligman & Verdery, 2011).

The institutional mechanism of collectivization combined legal coercion with direct physical violence. Peasants were formally required to sign voluntary adhesion forms joining the CAP, but the voluntariness of this process was systematically undermined by the secret police (*Securitate*) surveillance, arbitrary detention, confiscation of livestock and equipment, and the threat of criminal prosecution under the charge of sabotage of socialist construction (Iordachi & Dobrincu, 2009). The *chiaburi* – wealthy peasants defined loosely by landholding size and perceived class status – were subjected to disproportionate repression as the ideologically designated class enemy of collectivization, facing not only disproportionate arrest rates but also systematic exclusion from CAP membership, effectively expropriating their assets without any compensatory mechanism.[7]

[5]The Romanian Workers' Party Plenum of March 3–5, 1949 formally launched the collectivization campaign. See Iordachi & Dobrincu (2009), p. 27.

[6]Borșa (2013) provides a concise periodization of the campaign phases and their alignment with Soviet directives, noting that Romanian authorities adopted the Soviet model with minimal institutional adaptation.

[7]The legal definition of *chiabur* was deliberately kept imprecise, allowing local party activists to apply it opportunistically. See Kligman & Verdery (2011), pp. 113–119, for an analysis of how this definitional flexibility served coercive ends.

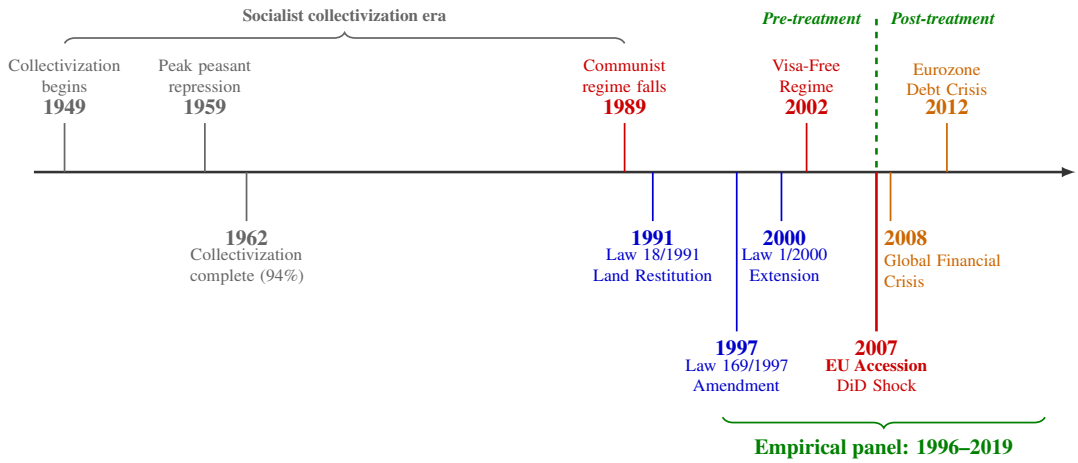


FIGURE 3: INSTITUTIONAL TIMELINE OF ROMANIAN AGRICULTURE, 1949–2019

Notes: Blue events denote domestic land restitution laws, while red and orange events highlight major structural and external macroeconomic shocks, respectively. The green bracket defines the empirical panel period used in the analysis, segmented by the 2007 EU Accession treatment shock.

The geographic variation in collectivization intensity is directly relevant to this paper's identification strategy. As Kligman and Verdery emphasize, the speed and intensity of the campaign varied considerably across regions, mediated by the interaction of local social structures, party activist constraints, and geo-historical conditions.[8] While Figures 4 and 5 display the speed and intensity variation across geography, Panel B of Table 1 quantifies it in collectivization rates (Kligman & Verdery, 2011) as well as arrests which specifically use data from Ivănescu (2015), who recovered detention figures from the *Securitate* archives for the peak repression year of 1959 in 16 regions at the time.[9] Here, the magnitude of the arrests in region r is estimated in this thesis with an index made using the following formula:

$$\text{ArrestIndex}_r = \frac{\text{PeasantsArrests}_r}{\text{RuralPopulation}_r} \times 10,000$$

weighting arrests per 10,000 rural inhabitants using population figures from the 1956 Romanian census (Suret-Canale, 1956). A parallel index is calculated for *chiaburi* arrests. The Plains regions record a mean collectivization rate of 57.4% by 1960, against 29.8% in the Mountains regions, a difference of nearly 28 percentage points which, despite inflated stan-

[8]Professor Gail Kligman, in correspondence with the author (November 2025), confirmed that this regional variability is a central analytical concern of Kligman & Verdery (2011) and directed the author to Iordachi & Dobrincu (2009) for detailed case study evidence on individual regions.

[9]Professor Dorin Dobrincu, in correspondence with the author (October 2025), confirmed that systematic county-level arrest data remains fragmentary and that Ivănescu's Securitate-based figures represent the most complete available source at the regional level. Dobrincu directed the author to his own documentary work for supplementary evidence: Dobrincu (2004).

dard deviation due to a small sample size, suggests that terrain ruggedness could be an eligible proxy for collectivization speed and intensity.

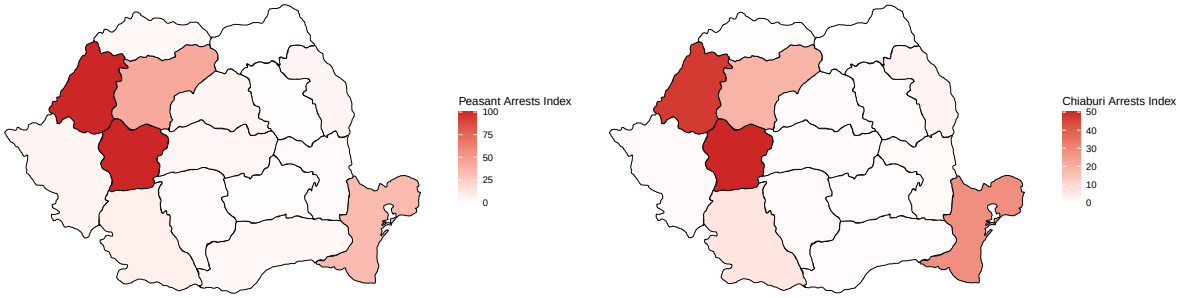
The arrest data, however, produces a counterintuitive pattern that merits careful interpretation. Mountain regions record a higher mean Peasants Arrests Index (74.7 versus 26.9 for the Plains) and a substantially higher *chiaburi* index (63.0 versus 11.1). This does not seem to indicate that collectivization was more intense in the Mountains in any productive sense – instead, it plausibly reflects the greater resistance capacity of Mountain communities, whose terrain afforded both physical refuge and stronger pre-existing social cohesion, forcing the state to deploy more coercive resources to achieve a lower final collectivization rate (Kligman & Verdery, 2011). In the Plains, the flat, exposed terrain and the legacy of large estate agriculture – which had already eroded smallholder solidarity – meant that resistance was structurally weaker and collectivization could be achieved at lower arrest intensity but higher ultimate coverage. The standard deviation on both arrest indices is extremely high, reflecting genuine regional heterogeneity rather than a uniform repressive pattern – a finding consistent with the regional variability emphasized by Kligman, Verdery, and Dobrincu.

What collectivization ultimately produced in the Plains was not merely a change in ownership structure but the construction of an entirely new physical infrastructure calibrated to industrial-scale agriculture: large-scale irrigation systems covering hundreds of thousands of hectares, centralized grain storage facilities, mechanized equipment pools, and processing plants integrated vertically within the CAP and IAS organizational structure.[10] This infrastructure was not incidental to the collective farm – it was constitutive of it, and its productive value was predominantly conditional on the scale and administrative integration of the socialist organizational form. The implications of this specificity for the post-1991 transition are developed later in this section.

Between 1949 and 1989, the Romanian state directed substantial capital investment into agricultural infrastructure in the Plains, concentrated in three areas: mechanization, irrigation, and centralized storage as well as processing capacity (Cartwright, 2017). Romania had no domestic tractor production at the end of the Second World War – the Brașov factory was converted from aircraft to tractor manufacturing only under Soviet occupation – and the subsequent expansion of the national tractor fleet from near zero to industrial scale within three decades was almost entirely absorbed by the CAP and IAS units of the Southern and Eastern Plains, where large-scale grain monoculture required mechanized tillage and harvesting at a scale impossible for individual smallholders.[11] Storage capacity was also centralized: each CAP operated collective grain depots and processing facilities integrated directly into state procurement chains, bypassing market mechanisms entirely.

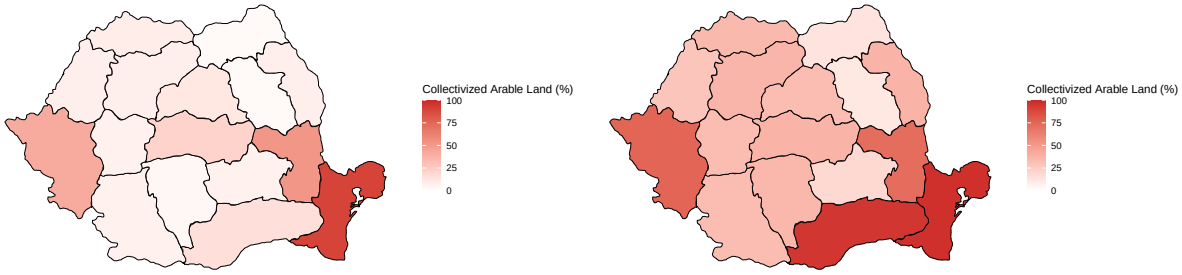
[10] By 1989, Romania's irrigation infrastructure covered approximately 3 million hectares, the vast majority concentrated in the Southern and Eastern Plains. See Urșanu et al. (2024), available at: https://www.sciencedirect.com/science/article/pii/S2213305424000262.

[11] On the Brașov factory conversion and the origins of Romanian tractor production under Soviet occupation, see Deletant, D. (2019) *Romania under Communism: Paradox and Degeneration.* London: Routledge, pp. 89–92.



(a) Total Peasants' Arrests (b) Total *Chiaburi* Arrests

FIGURE 4: SPATIAL VARIATION IN PEASANT ARRESTS



(a) Collectivization Rates Pre-Arrests (b) Collectivization Rates Post-Arrests

FIGURE 5: SPATIAL VARIATION IN COLLECTIVIZATION SPEED (1958-1960)

In the Mountains, the collectivization period did not produce as much equivalent capital accumulation. The terrain precluded large-scale irrigation – the CAP units established there were smaller, less mechanized, and organized around livestock rather than grain, meaning the capital stock they operated – barns, small equipment, pasture management infrastructure – was closer in character to what had existed before collectivization than to the industrial apparatus constructed in the Plains. The aforementioned Mathijs and Swinnen (1998) document this contrast systematically across Central and Eastern European transition economies, finding that regions entering the transition with more capital-intensive, irrigation-dependent agricultural systems faced structurally larger adjustment costs precisely because their inherited assets were less redeployable at the household scale.

TABLE 1: HISTORICAL ARRESTS AND COLLECTIVIZATION (1959–1960)

	Obs.	Mean	Std. Dev	Min	Median	Max
Panel A: Full Sample Averages						
Peasant Arrests Index	16	50.79	135.85	0.05	4.4	538.79
Chiaburi Arrests Index	16	37.01	118.94	0.05	1.285	480.33
Collectivization Rate	16	43.56	26.61	12.0	35.5	97.0

	Plains (TRI < Median)			Mountains (TRI > Median)		
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev
Panel B: Comparison by Geography						
Peasant Arrests Index	8	26.9	53.5	8	74.7	188.0
Chiaburi Arrests Index	8	11.1	17.2	8	63.0	169.0
Collectivization Rate	8	57.4	31.2	8	29.8	10.5

Panel C: Regional Classification by TRI Score

Plains (TRI < Median)		Mountains (TRI > Median)	
Region	TRI Score	Region	TRI Score
Bucharest	4.5	Suceava	32.8
Dobrogea	9.1	Maramureș	33.1
Galați	14.2	Brașov	36.0
Iași	19.7	Argeș	36.6
Oltenia	20.3	Bacău	37.8
Crișana	22.7	Maghiară	39.9
Banat	26.6	Cluj	44.5
Ploiești	31.4	Hunedoara	54.1

Notes: Arrest data refers to the year 1959; Collectivization rates refer to 1960. Panel C lists the 16 historical regions sorted by their median Terrain Ruggedness Index (TRI). 'Plains' indicates TRI below the sample median; 'Mountains' indicates TRI above the median.

C. Restitution & EU Accession

The infrastructure was inherited by the Romanian state in December 1989, but the legal framework designed to dismantle socialist agriculture treated it as though it did not exist. **Law 18/1991**, the foundational land restitution legislation passed by the Romanian parliament on February 19, 1991, established the formal mechanism for returning collectivized land to its pre-communist owners or their heirs.[12] The law mandated restitution of up to 10 hectares per family in the original plot configurations – that is, in the strip parcels that had existed prior to collectivization – without any consolidation mechanism, without any provision for the col-

[12] *Legea fondului funciar nr. 18/1991*, published in *Monitorul Oficial al României, Partea I, nr. 37, 20 februarie 1991*. The full text is available through the Romanian parliament's legislative database at https://legislatie.just.ro/Public/DetaliiDocument/280373.

lective management of inherited infrastructure, and without any compensation framework for assets that could not be meaningfully individualized.[13] Irrigation canals, machinery depots, storage silos, and processing facilities – the indivisible collective infrastructure documented in the previous paragraphs – fell outside the restitution framework, passing nominally to local authorities or newly formed commercial companies, but without the financial resources, technical capacity, or legal clarity required to maintain them (Verdery, 2003).

The consequences were immediate. Without a functioning operator for the irrigation infrastructure, maintenance ceased and the network deteriorated rapidly: World Bank estimates from the mid-1990s suggest that effective irrigated area in Romania fell from approximately 3 million hectares at the end of the communist period to under 1 million hectares by 1995, a collapse of more than two thirds within five years of restitution.[14] Machinery pools were either auctioned at distressed prices to private buyers who lacked the scale to deploy them productively, or simply abandoned. The result was the dead capital configuration discussed in the literature review (De Soto, 2001): assets with significant replacement value but negligible productive value in the new property rights environment.

Verdery (2003) captures the political logic underlying this design with particular precision – land restitution in Romania was conceived primarily as a symbolic act of historical justice and a repudiation of communist expropriation, rather than as an economic policy aimed at productive restructuring. The consequence was the aforementioned "fuzzy property" – formal ownership rights that were legally ambiguous, spatially contested, and institutionally unsupported, discouraging both consolidation and investment. Cartwright (2017) reinforces this interpretation, documenting how the restitution commissions established under Law 18/1991 were systematically under-resourced, politically manipulated, and technically incapable of resolving the competing claims that strip restitution inevitably generated.[15]

Subsequent legislative attempts to address these failures were partial and belated. **Law 169/1997** amended the restitution framework to extend coverage and correct some procedural irregularities, and **Law 1/2000** further expanded restitution to include forest land and raised the ceiling to 50 hectares.[16] Neither law introduced a consolidation mechanism or resolved the governance vacuum over collective infrastructure. The fundamental asset mismatch – fragmented private land ownership coexisting with orphaned collective capital – remained structurally

[13] Article 8 of Law 18/1991 explicitly stipulated restitution in the original location (*pe vechiul amplasament*) where physically possible, a provision that directly produced strip fragmentation in the Plains where pre-collectivization plots had been narrow and elongated. See Cartwright (2017) for a detailed legal analysis of this provision and its practical consequences.

[14] World Bank Integrated Water Resources Rapid Assessment Report of 2013: Romania. *Washington D.C.: World Bank*. Available at: https://documents1.worldbank.org/curated/en/675851468094456585/pdf/842640WP0P14660Box0382136B00PUBLIC0.pdf.

[15] Cartwright (2017) also documents numerous cases in which restitution commissions awarded overlapping claims to the same plot, leaving ownership legally unresolved for years and effectively preventing any productive use of the land.

[16] *Legea nr. 169/1997 pentru modificarea și completarea Legii fondului funciar nr. 18/1991, Monitorul Oficial nr. 299/1997; Legea nr. 1/2000 pentru reconstituirea dreptului de proprietate asupra terenurilor agricole și celor forestiere, Monitorul Oficial nr. 8/2000.*

intact throughout the 1990s and into the 2000s, and it is this configuration that the empirical analysis of this paper inherits as its baseline condition.

Since the 1991 restitution legislation, Romania's accession to the European Union on January 1, 2007 represented the most significant external institutional shock to the agricultural sector, introducing three simultaneous changes whose distributional consequences across terrain types were far from uniform.

The first and most immediate change was the activation of Common Agricultural Policy direct payments. Romanian farmers became eligible for the Single Area Payment Scheme (SAPS), a simplified per-hectare subsidy disbursed regardless of production decisions.[17] Because SAPS payments are proportional to registered land area rather than to productive output or capital investment, they disproportionately benefited large landholders and corporate farm operators who had consolidated significant acreage in the Plains during the 1990s – not the smallholder households that restitution had created. Rizov et al. (2013) find that pre-decoupling CAP subsidies negatively affected farm total factor productivity across EU member states, suggesting that subsidy access may have reduced the pressure to restructure rather than facilitating it.

The second change was the activation of single market labor mobility rights. Romanian citizens gained the right to work legally across most EU member states from 2007 onward, accelerating an emigration process that had already begun through a visa-free regime started in 2002 as well as through bilateral agreements with Spain and Italy in the early 2000s.[18] The critical distributional question is whether accession-era emigration was accessible equally across terrain types. Sandu (2005) establishes that migration from Romania is heavily network-dependent, with transnational migration chains concentrated in communities that had developed pioneer migrants in earlier waves – communities located disproportionately in the Mountains and in western regions with stronger pre-existing connections to Western European labor markets. Plains communities, whose spatial immobility had prevented the accumulation of transnational social capital during the 1990s, faced structurally higher barriers to participating in post-accession labor mobility, although the relationship is heterogeneous rather than deterministic.

Finally, the third change was the requirement to implement EU rural development regulations under the SAPARD pre-accession instrument and subsequently the EAFRD framework. It conditioned structural funds on administrative capacity and co-financing ability that Plains counties were systematically less equipped to mobilize despite also receiving subsidies through other EU-financed frameworks.[19]

[17] Romania adopted SAPS rather than the standard CAP payment entitlement system, a transitional arrangement permitted for new member states under Council Regulation (EC) No. 1782/2003. SAPS payments were phased in gradually, reaching 100% of the standard rate by 2016.

[18] Spain and Italy, the two primary destinations for Romanian labor migrants, opened their labor markets to Romanian workers progressively from 2002 onward through bilateral agreements, predating formal accession. See Sandu (2005) for the pre-accession migration network structure.

[19] Council Regulation (EC) No. 1268/1999 established the SAPARD instrument for pre-accession rural development; it was succeeded by the European Agricultural Fund for Rural Development (EAFRD) under Council Regulation (EC) No. 1698/2005 upon accession.

Naturally, these three simultaneous changes did not operate in a neutral institutional environment. In the Plains, EU accession intersected with an already-consolidated political economy in which ex-communist elites with disproportionate access to land and mechanized capital had established clientelist relationships with PSD-controlled local administrations, collectively generating the bureaucratic barriers and resource hoarding that suppressed independent business formation. Figure 6 schematizes the resulting equilibrium: sub-market land leases, state contracts channeled through political loyalty, and administrative barriers to entry jointly crowded out the independent business sector that accession-era market integration might otherwise have stimulated. Hence, taking the three simultaneous transformations and the institutional context into account, the 2007 becomes a methodologically attractive shock for identification purposes: it was externally determined, applied uniformly in legal terms across all Romanian counties, and its timing was sufficiently abrupt to generate pre- and post-accession dynamics. Crucially, because the shock operated through channels – subsidy access, labor mobility, administrative capacity – whose distributional consequences were conditioned by the pre-existing asset mismatch and the political capture structure, it functions as an intensifier of divergence rather than an independent treatment, which is precisely what the event study design in Section VI exploits.

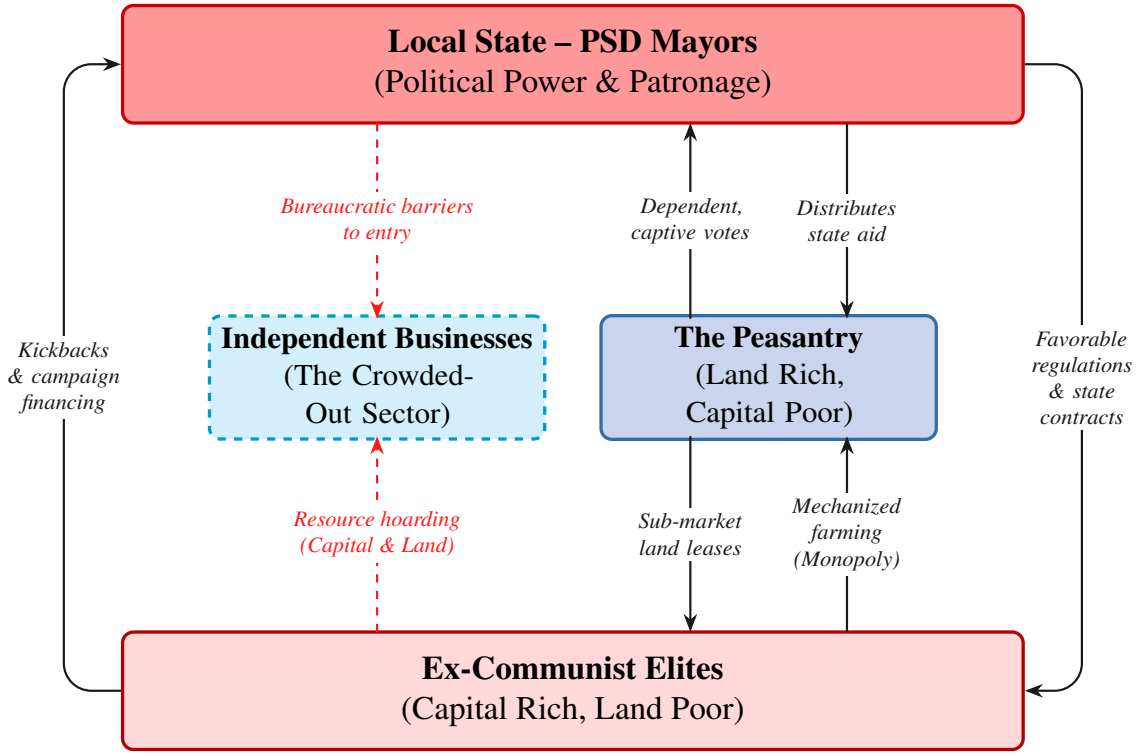


FIGURE 6: RESOURCE EXTRACTION AND BUSINESS SUPPRESSION IN THE PLAINS

Notes: Solid arrows denote self-sustaining resource flows between the state, economic elites, and the dependent peasantry. Dashed red arrows indicate the structural, exclusionary pressures actively weakening the independent business sector.

IV. Conceptual Framework

The institutional shock of 1991 restitution did not affect Romanian households uniformly – its consequences were mediated by the nature of the capital stock that collectivization had constructed, which was itself geographically determined by terrain. This section maps the institutional constraints established in Section III onto the decision environment facing individual actors – peasant households, potential entrepreneurs, the elites, and local politicians – and derives the county-level observables that each actor's rational response generates, which are summarized in Table 2.

A Plains household receiving restituted land in 1991 faced a constrained choice set. With strip parcels too small for viable mechanized grain cultivation, no access to the irrigation networks or machinery pools that had made parcels productive under the CAP, and insufficient liquid income to finance either productive investment or relocation, the rational response was subsistence farming – not because households preferred it, but because the asset mismatch eliminated the alternatives. Selling or leasing land was complicated by legally unresolved ownership claims and the absence of functioning land markets – migration required upfront resources that subsistence plots could not generate. This behavioral lock-in is what eventually produced the tractor deficit: Plains counties are predicted to exhibit systematically lower tractor density not because terrain precludes mechanization, but because the hold-up problem destroyed the incentive to invest in replacement capital after the vertically integrated CAP structure dissolved (Gow & Swinnen, 1998). Tractors occupy a specific analytical position here – as collective CAP property disposed of separately from land under Law 18/1991, their post-1991 trajectory is determined solely by market investment decisions, making tractor density a clean measure of productive restructuring uncomplicated by the legal ambiguities that surround land privatization itself. Plains counties are therefore predicted to generally show lower tractor density unconditionally, with the deficit widening after 2007 as EU accession accelerated modernization in counties capable of restructuring.

Privatization rate is predicted to be lower in the Plains for the same underlying reason: where complementary capital is absent and property claims legally contested, private operation is unattractive and privatization stalls. This makes privatization rate a revealed measure of asset mismatch severity – an outcome in its own right, but also a mediator through which the Plains dummy might operate on some downstream variables. It is included as a right-hand side control in the OLS specifications for remaining outcomes precisely to separate the direct Plains effect from the portion that flows through delayed privatization.

The predictions for rural business formation are conditional rather than unconditional, and this conditionality is an important analytical puzzle of the paper. A potential entrepreneur evaluating entry into a Plains county faces two countervailing forces. On the one hand, Plains counties offer better road access, flatter terrain, lower transport costs, and greater market proximity – structural advantages that should favor non-farm business formation independently of agricultural performance. On the other hand, in counties where political capture is deeper, the local PSD administration generates bureaucratic barriers to entry, channels state contracts through political loyalty, and tolerates elite monopolization of mechanized farming and land

leasing that raises the effective cost of independent market entry (see Figure 6). The net prediction could then be the following: unconditionally positive or at least insignificant Plains effect on business formation and a negative or smaller magnitude effect when conditioned on high PSD vote share. Low-PSD Plains counties, freed from the capture equilibrium, are predicted to show larger business formation responses to post-2007 market integration.

The demographic and fiscal predictions follow a similar moderating logic. A local politician governing an immobile, transfer-dependent electorate faces lower costs of clientelist commitment relative to programmatic competition – the exit option that would otherwise discipline political actors is almost unavailable. This behavioral incentive sustains fiscal dependency on central transfers and suppresses local revenue generation, but the effect is predicted to be visible only in High-PSD Plains where the capture equilibrium is operative. Rural natural increase is similarly predicted to decline most severely in High-PSD Plains, where economic stagnation is deepest and demographic immobility most entrenched.

External migration is a theoretically ambiguous outcome. Pre-2007, the liquidity constraint suppresses migration unconditionally by preventing households from covering relocation costs. Post-accession labor mobility rights reduce this barrier, but heterogeneously – Mountain communities with established transnational pioneer networks exploit the shock faster, while Plains communities face compounding barriers from both the liquidity constraint and the absence of migration networks (Sandu, 2005). Hence, the net Plains effect on external migration might just be indeterminate in sign, and the analysis treats it as exploratory, with primary interest in whether post-2007 dynamics differ between High- and Low-PSD Plains rather than in the unconditional level effect.

TABLE 2: TESTABLE IMPLICATIONS OF THE CONCEPTUAL FRAMEWORK

Outcome Variable	**Unconditional Plains Effect**	**Conditional on High PSD Capture**	**Primary Timing**
Asset Mismatch Channel			
Privatization Rate	$-$	*Not Tested*	1996–1999
Tractor Density	$-$	$--$	Post-2007
Immobility & Political Capture Channel			
Rural Business Formation	$+$ / 0	$-$ / 0	Post-2007
External Migration	$\pm$	$\pm$	Post-2007
Rural Natural Increase	$-$ / 0	$--$	Post-2007
Fiscal Dependency	0	$+$	Post-2007

Notes: $-$ and $+$ denote the predicted direction of the Plains dummy coefficient. Double signs ($--$, $++$) indicate a stronger predicted magnitude conditional on high PSD vote share. 0 indicates no predicted unconditional level effect. $\pm$ indicates theoretical ambiguity in sign. PSD Vote Share serves as the moderating variable in split-sample and Synthetic DiD specifications rather than as an outcome variable. Timing refers to the period in which divergence is predicted to be most pronounced relative to the 2007 EU accession shock.

V. Data

Empirical analysis is based on an unbalanced cleaned panel of 40 Romanian counties (*județe*) that originates from the full dataset of 42 counties. However, two counties are excluded from the panel throughout: Bucharest and Ilfov. Both are classified as Plains by terrain ruggedness, but their economic structure, fiscal arrangements, and demographic dynamics are categorically determined by urban agglomeration effects rather than agricultural history – including them would thus introduce a systematic comparability problem for all outcome variables whose variation is premised on rural economic structure.

The panel covers the period 1996-2019, with the unit of observation being the county-year. It therefore covers 19 Plains counties and 21 Mountain counties across 24 years, yielding a maximum of 960 county-year observations per variable, with variation in coverage arising from data availability across specific outcomes documented in the descriptive statistics – for example, variables such as business formation and hospital bed density have 920 observations due to unavailable data in 1997, whereas rural natural increase lacks one observation for Neamț in 1997. All economic, demographic, and administrative variables and their components are sourced from the Romanian National Institute of Statistics (INS, *Institutul Național de Statistică*) with the help of its online database and its statisticians.[20] Electoral data are sourced from the Permanent Electoral Authority (*Autoritatea Electorală Permanentă*).[21]

Our main treatment variable is a binary Plains dummy constructed from the Terrain Ruggedness Index (TRI), calculated directly in *QGIS* from a 90-meter resolution digital elevation model sources from the Shuttle Radar Topography Mission (SRTM).[22] TRI was computed using the QGIS Terrain Analysis plugin, applying the Riley et al. (1999) algorithm which defines ruggedness as the mean absolute difference in elevation between a focal cell and its eight surrounding neighborhoods. County-level TRI values represent the spatial mean of all raster cells falling within each county boundary. The index thus captures local relief variation rather than absolute elevation, which more precisely proxies the flat, irrigable lowland versus rugged, fragmented highland distinction that historically determined agricultural collectivization suitability – a flatter county is not necessarily low-lying, but it is one whose terrain permitted large-scale mechanized farming. Counties with TRI below the sample median are classified as Plains whereas those above as Mountains. While the exact TRI means are listed in the Appendix A, Figure 1 already visualized the index spatially, whereas Figure 7 additionally presents the distribution following the separation by the median. The latter produces a treatment group of 19 counties (after excluding Bucharest and Ilfov) whose mean TRI score is no higher than 26.3, against a control group of 21 Mountain counties whose mean TRI score is 26.5 or more. In order to verify that the TRI distribution constitutes a genuine geographic

[20] We are particularly thankful to Ms. Marican and Mr. Vîrdol who have assisted in obtaining the crucial datasets on county-level agricultural outcomes. Other variables were sourced from the online database available at: http://statistici.insse.ro/tempoins/?page=search&lang=en.

[21] Data was compiled with the help of Commit Global platform. Available at: https://istoric.rezultatevot.ro/elections/60/results.

[22] SRTM elevation data were obtained from the Consortium for Spatial Information (CSI) database. Available at: https://srtm.csi.cgiar.org/srtmdata/.

discontinuity rather than a boundary effect, Section VIII will test this directly in the extreme TRI robustness check.

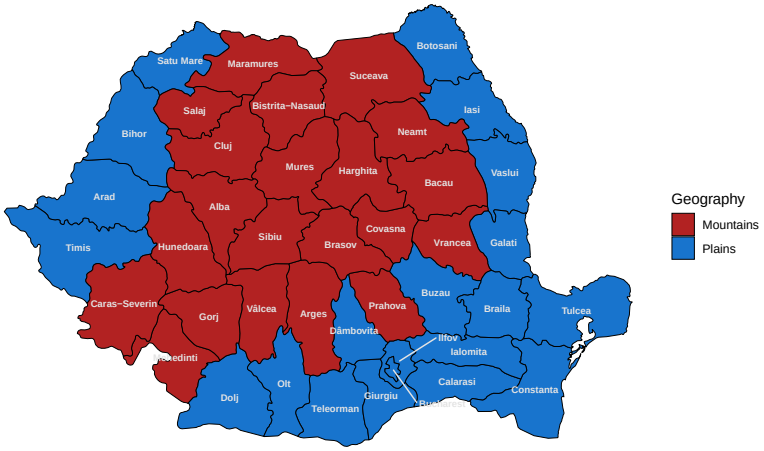


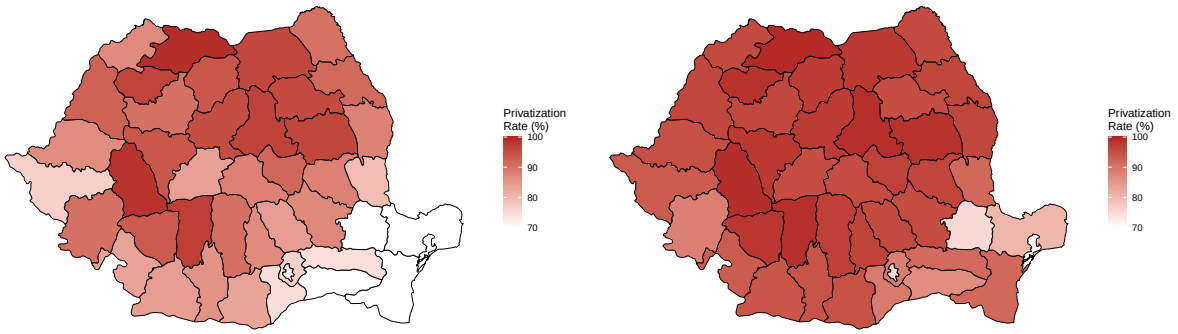
FIGURE 7: PLAINS AND MOUNTAINS DIVISION

Notes: Division is determined by the Median TRI value.

A central measure of decollectivization intensity, privatization rate, is measured as the percentage of agricultural land privatized over the period 1996-2014, sourced from INS land registry records. Here, the 1996-2014 period is the largest possible time window provided as data is no longer reported after 2014. The longer panel is preferred over a shorter early-transition measure because the paper's interest is the cumulative pace of privatization over the medium run rather than the initial shock – a county that privatized slowly throughout the 1990s and 2000s is one where the governance vacuum over collective infrastructure persisted longest and it is this persistence that the variable is intended to capture. As discussed in Section IV, privatization rate serves a dual role in the empirical strategy first as an outcome in the OLS baseline and then as a right-hand side mediator control in the OLS specifications for the remaining outcomes in order to isolate the Plains effect net of the portion that functions through delayed privatization. Its spatial variation, similar to that of the TRI, is displayed in Figure 8.

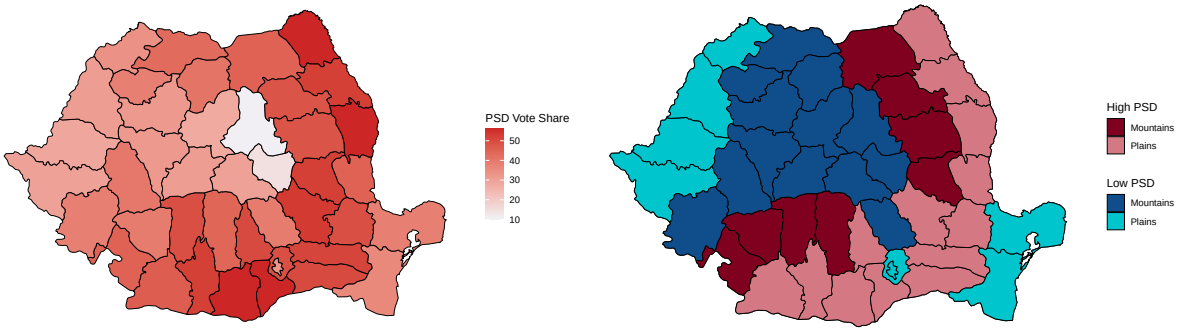
PSD vote share, an endogenous but treatment-relevant variable, is measured as the percentage of valid votes cast for the Social Democratic Party in the first round of Romanian legislative elections in order to isolate party-specific loyalty from individual-candidate-specific factors and protest votes. Legislative election results at the county level are available from the Permanent Election Authority for the election years 1996, 2000, 2004, 2008, 2012, and 2016. The 2012 election is excluded from the PSD vote share variable due to the party running under a different coalition label that year, which would introduce a mechanical discontinuity in the series. Because elections occur every four years, the OLS specification uses observations only for those years during which elections were held. For the split-sample difference-in-differences and Synthetic difference-in-differences specifications, PSD vote share functions

as a moderator rather than an outcome: each county's mean PSD share across all available elections is computed, and counties above the sample median of this time-invariant mean are classified as High-PSD, those below as Low-PSD (Figure 9). This partition is designed to detect structural political capture – the stable, decade-spanning dominance of clientelist networks – rather than electoral volatility in any single cycle. In the OLS baseline, PSD vote share also appears as an outcome variable to establish the geographic baseline association between Plains status and political capture.



(a) Mean Privatization 1996-1999 (b) Mean Privatization 1996-2014

FIGURE 8: PRIVATIZATION ACROSS TIME



(a) PSD Vote Share by County (b) Political Plains and Mountains Separation

FIGURE 9: SPATIAL VARIATION OF POLITICAL CAPTURE

Notes: Division is determined by both the Median TRI and Median PSD Vote Share value.

The primary outcome variable for the asset mismatch channel is tractor density, measured as the number of tractors per 100 hectares of arable land at the county level.[23] The normalization by arable land rather than total county area or population is deliberate – the asset mismatch mechanism predicts a deficit in agricultural capital investment specifically, and arable land is the relevant denominator for measuring whether the capital stock is adequate for the farming activity the terrain supports. The cleaned sample excludes two extreme outliers like Ilfov and Bucharest, which, in particular, has its tractor density more than ten times larger than the rest of the counties. Its spatial variation and that of some other outcome variables can be found in the Appendix B of this paper.

The remaining four outcome variables capture the downstream demographic and political consequences of the asset mismatch and political capture mechanisms. Rural business formation measures rural firm registrations per 1,000 permanent rural inhabitants, sourced from INS business registry data. The rural restriction – firm registrations in localities classified as rural by Romanian administrative law – is key: urban business formation within Plains counties would conflate the agricultural asset mismatch mechanism with urban agglomeration effects unrelated to collectivization history. To explicitly capture the political mechanism of the institutional trap, the fiscal dependency ratio measures the percentage of central state subsidies over total local revenues. High levels of fiscal reliance serve as a proxy for a county's vulnerability to the clientelist resource allocation of the PSD machine.

As far as demographics are concerned, rural natural increase measures the balance of births and deaths per 1,000 rural inhabitants. The rural restriction here follows a similar logic as for businesses – the demographic mechanism operates through immobility in agricultural communities specifically, and total natural increase would dilute this signal with urban dynamics. Finally, external migration rate measures net international migrants per 1,000 inhabitants calculated using the following formula:

$$\text{Net Migration Rate}_{c,t} = \frac{\text{Arrivals}_{c,t} - \text{Departures}_{c,t}}{\text{Population}_{c,t}} \times 1,000$$

Intuitively, negative values of this measure indicate net emigration and vice versa. The only limitation of this metric is that it is measured at the total county level rather than the rural population specifically, as Romanian administrative records do not disaggregate international migration flows by rural-urban origin – a limitation shared across the economics literature on Romania (Sandu, 2005). Nevertheless, the spatial variation of the total measure, especially its evolution overtime, seems like a reliable indicator of the immobility trap and the liquidity constraint following shocks such as visa-free regime announcement and EU Accession that coincide with larger international emigration in the Mountains and less so in the Plains (Figure 10).

[23] Tractor stock and arable land area data are sourced from INS. The per-hectare normalization is preferred over a per-capita measure because the mechanism operates through agricultural capital intensity rather than population density — a county with few tractors relative to its arable area is one that has failed to mechanize its agriculture, regardless of how many people live there.

All specifications, except the Synthetic DiD one, include four time-varying socioeconomic controls selected to account for pre-existing county-level development differences that could confound the Plains-Mountains comparison. Modernized roads ratio measures the percentage of modernized and light asphalt roads in the total county road network, also sourced from the INS, and proxies for the quality of transport infrastructure – a county with better roads might show different business formation and fiscal dependency independently of its agricultural history. High school graduates ratio measures the number of high school graduates per 1,000 inhabitants and controls for human capital differences that could independently affect both migration and business formation outcomes. Hospital beds index measures the number of hospital beds per 1,000 inhabitants and indicates public service provision quality – a control for the possibility that fiscal dependency reflects genuine service delivery differences rather than clientelist capture. Non-political social aid index is measured by weighting aggregate spending on unemployment indemnities by the permanent resident population in a county. It is specifically chosen over broader social transfer measures because it captures automatic, eligibility-based transfers rather than discretionary political allocations – the latter would be endogenous to the political capture mechanism the paper is testing.

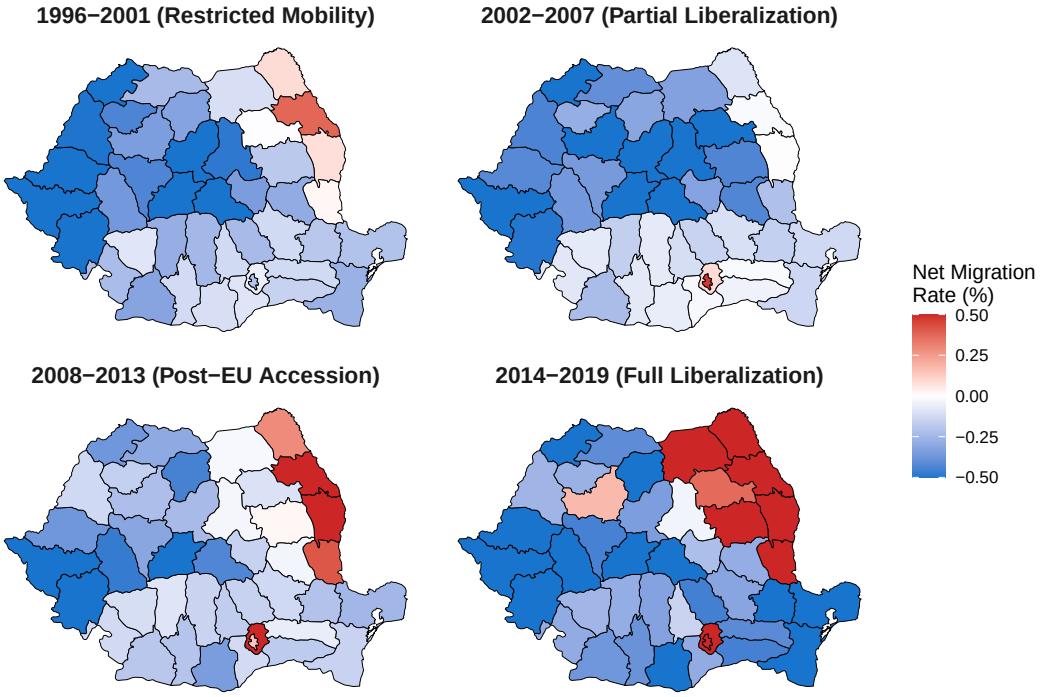


FIGURE 10: INTERNATIONAL NET MIGRATION OVER TIME

Notes: Positive values entail more immigration than emigration and vice versa.

Table 3 presents summary statistics for the cleaned sample while the pre-processed descriptive statistics can be found in Appendix A. Panel A of the table reports on the full sample whereas Panel B provides a Plains-Mountains comparison with t-tests for mean differences. Several patterns in Panel B are directly relevant to the identification strategy and warrant attention before the empirics. First, the tractor deficit is large and precisely estimated: Plains counties average 1.71 tractors per 100 arable hectares against 3.86 in the Mountains, a difference significant at the 1% level – a raw descriptive gap that the regression analysis then attributes causally to the asset mismatch mechanism. Second, the privatization gap is similarly significant at 1% no matter what time frame is being taken which is consistent with the predicted effect of the governance vacuum over collective infrastructure on the pace of land market development. Third, rural business formation seems to run in the opposite direction, with Plains having more firms per 1,000 inhabitants than the Mountains, significant at 5%, confirming that the unconditional Plains effect is positive and that suppression in the Plains might only be recoverable conditional on PSD capture intensity.

Net migration, although confirmed that it is higher in the Mountains enforcing the immobility trap theory in the Plains, does not indicate a significant difference after a t-test which validates the guess about ambiguous effects on migration following institutional shocks. Insignificant difference for rural natural increase and fiscal dependency motivates the need to delve into the political capture mechanism in the identification strategy, too. At the same time, PSD vote share being significantly higher in the Plains counties establishes the geographic baseline association between terrain and political capture. As for controls, the most notable differences are in high school graduation rates and the social aid index – significantly higher in the Mountains –, reflecting the historically stronger non-political welfare infrastructure in Transylvanian counties. Hence, inserting them in regression specifications should aid in preventing omitted variable bias. Nevertheless, while road quality and hospital beds density show no significant unconditional difference, including them should instead help absorb residual variance.

TABLE 3: DESCRIPTIVE STATISTICS

	Obs.	Mean	Std. Dev	Min	Median	Max
Panel A: Cleaned Sample						
Core Variables						
Privatization Rate (1996-1999)	160	86.60	10.40	58.60	88.20	99.30
Privatization Rate (1996-2014)	760	94.40	7.38	58.60	97.50	99.80
Tractor Density	960	2.84	2.16	0.46	2.16	15.58
Modern Variables						
Rural Business Formation	920	1.55	0.85	0.33	1.35	4.79
Net Migration Rate	960	-0.03	1.82	-3.25	-0.23	23.01
Rural Natural Increase	959	-4.41	3.76	-14.45	-4.35	5.86
Fiscal Dependency Ratio	960	13.60	9.54	0.04	11.54	47.49
PSD Vote Share	200	35.09	14.70	4.00	34.50	69.00
Control Variables						
Modernized Roads Ratio	960	57.94	10.87	33.45	57.37	86.04
High School Grads Ratio	960	10.63	2.79	4.10	10.34	20.57
Hospital Beds Index	920	5.84	1.58	2.62	5.58	11.11
Social Aid Index	960	32.80	30.66	0.48	23.80	237.13

	Plains (TRI < Median)			Mountains (TRI > Median)			t-test
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev	Value
Panel B: Comparison by TRI Score							
Core Variables							
Privatization Rate (1996-1999)	76	80.10	9.91	84	92.40	4.83	4.89***
Privatization Rate (1996-2014)	361	92.00	5.71	399	96.60	2.48	3.28***
Tractor Density	456	1.71	0.78	504	3.86	1.71	5.21***
Modern Variables							
Rural Business Formation	437	1.81	0.83	483	1.31	0.52	-2.28**
Net Migration Rate	456	0.24	1.37	504	-0.29	0.38	-1.64
Rural Natural Increase	456	-5.09	3.69	503	-3.78	3.35	1.17
Fiscal Dependency Ratio	456	13.90	1.74	504	13.40	2.20	-0.69
PSD Vote Share	95	39.70	9.74	105	30.90	11.60	-2.49**
Control Variables							
Modernized Roads Ratio	456	59.30	9.34	504	56.70	8.02	-0.93
High School Grads Ratio	456	10.10	1.17	504	11.10	1.02	3.02***
Hospital Beds Index	437	5.50	1.42	483	6.15	1.39	1.45
Social Aid Index	456	26.40	4.90	504	38.70	13.40	3.94***

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Panel A reports summary statistics for the cleaned sample (1996–2016). Panel B divides the sample by the median Terrain Ruggedness Index (TRI). 'Plains' indicates TRI below the sample median; 'Mountains' indicates TRI above the median.

VI. Identification Strategy

The empirical analysis proceeds in three stages to identify the legacy of Romania's agricultural decollectivization. First, baseline OLS panel regressions establish structural correlations between the geographic asset mismatch (the Plains), privatization rates, and rural outcomes. To address unobserved time-invariant heterogeneity, the analysis then employs a dynamic Event Study leveraging the exogenous shock of the 2007 EU Accession. Finally, to isolate the mechanism of political capture, the strategy applies Split-Sample and Synthetic Difference-in-Differences (SDiD) estimators (Arkhangelsky et al., 2021). Across all models, causal identification relies fundamentally on the strict exogeneity of the geographic treatment (terrain ruggedness) and the Stable Unit Treatment Value Assumption (SUTVA) – presuming no systemic spillovers from the Plains to the Mountains control group.

A. Ordinary Least Squares

The analysis begins by estimating the persistence of the agricultural asset mismatch across a standard cleaned panel of 40 counties. To first establish that geographic fundamentals inherently influenced the speed of economic transition, the annual privatization rate (Pr_{ct}) is regressed directly on the geographic treatment, conditioning on initial baseline disparities:

$$Pr_{ct} = \alpha_0 + \rho \cdot Pl_c + \gamma_1 \cdot \mathbf{X}_{c,1996} + \delta_t + \varepsilon_{ct} \tag{1}$$

Next, the primary structural OLS specification evaluates how this geographic and institutional mismatch translates into downstream rural and political outcomes:

$$Y_{ct} = \alpha_1 + \beta_1 \cdot Pl_c + \beta_2 \cdot Pr_{ct} + \gamma_2 \cdot \mathbf{X}_{c,1996} + \delta_t + \varepsilon_{ct} \tag{2}$$

where Y_{ct} represents the outcome of interest such as tractor density, migration, and others in county c at time t. The variable Pl_c is a time-invariant binary treatment indicator taking the value of 1 if the county's TRI falls below the sample median, capturing the flatlands susceptible to the capital-land mismatch. Pr_{ct} tracks the rate of agricultural land privatization. To absorb aggregate macroeconomic shocks affecting all counties – such as the 1999 recession or 2007 EU Accession – both models include year fixed effects (δ_t).

A critical identification choice in both specifications is the inclusion of $\mathbf{X}_{c,1996}$, a vector of socioeconomic covariates like modernized roads, high school graduation rates, and others. They are strictly fixed at their 1996 baseline levels.[24] Including time-varying controls in regressions evaluating a long-term transition process risks introducing post-treatment bias ("bad controls"). Because infrastructure, education, and welfare naturally evolve as consequences

[24] Here, the standard panel utilizes N=760 annual county-years because privatization rates contain data only until 2014, while the PSD Vote Share regressions rely on N=160 county-election years. The Rural Natural Increase model operates with N=759 due to a missing historical observation for Neamț in 1997, and Rural Business Formation utilizes N=720 as business registry data begins post-1996.

of a region's economic trajectory and privatization speed, controlling for their contemporaneous values would artificially absorb the exact mechanisms this paper measures. Fixing these controls helps isolate the geographic penalty from historical baseline differences – such as the stronger institutional infrastructure in Transylvanian Mountain counties – without over-controlling for the downstream effects of the policy shock.

Finally, to account for serial correlation within regions over time, standard errors are clustered at the county level. The main text reports the full specifications, while unadjusted baseline models (without controls) are provided in Appendix D.

B. Baseline Difference-in-Differences & Dynamic Event Study

To move from structural correlations to causal identification, the analysis leverages the exogenous macroeconomic shock of Romania's 2007 EU Accession. By gradually lifting mobility barriers starting in 2002 and abruptly removing trade restrictions in 2007, EU integration exposed the latent vulnerability of the Plains' agricultural mismatch. Capturing the Average Treatment Effect on the Treated (ATT) across the entire post-accession period will require a standard Two-Way Fixed Effects (TWFE) model to be estimated:

$$Y_{ct} = \beta \cdot (Pl_c \times Post_t) + \gamma \cdot \mathbf{X}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \tag{3}$$

where Y_{ct} again denotes the outcome for county c at time t, $Post_t$ is an indicator taking the value of 1 for the years 2007 onward, and β captures the pooled divergence. To observe the precise temporal evolution of this divergence and visually verify the parallel trends assumption, this framework is expanded into a dynamic event study:

$$Y_{ct} = \sum_{k \neq 2006} \beta_k \cdot (Pl_c \times \mathbf{1}_{t=k}) + \gamma \cdot \mathbf{X}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \tag{4}$$

where the dynamic parameters of interest, β_k, estimate the annual gap between the Plains and Mountains relative to the pre-treatment base year of 2006.

A critical methodological change here is the inclusion of county fixed effects (α_c). In the baseline OLS model, county fixed effects had to be omitted because they would perfectly absorb the time-invariant geographic treatment (Pl_c). In this panel framework, however, the treatment effect is isolated through the interaction term. This allows α_c to rigorously control for all unobserved time-invariant heterogeneity such as historical culture and permanent soil quality without absorbing the divergence of interest. Year fixed effects (δ_t) are retained to absorb aggregate national shocks like the 2008 global financial crisis.[25]

This fixed-effects structure also necessitates a shift in the control vector ($\mathbf{X}_{ct}$). Given that the static 1996 controls used in the OLS regressions are perfectly collinear with α_c, time-varying controls are introduced to account for localized developmental trajectories. To address

[25]The panel utilizes N=960 annual county-years. Specifications incorporating the Hospital Beds Index or Rural Business Formation utilize slightly reduced panels (N=920 or 919) due to data availability constraints.

the standard econometric risk that these time-varying covariates might act as endogenous "bad controls" following the EU shock, all models are estimated both with and without $\mathbf{X}_{ct}$ (Appendix D) to prove the treatment effect is not an artifact of post-treatment conditioning.

As mentioned before, causal interpretation here strictly relies on the parallel trends assumption. The event study (Equation 4) directly tests this by visualizing the pre-trends. Standard errors are again clustered at the county level to correct for arbitrary serial correlation.

C. Split-Sample Difference-in-Differences

While the baseline models isolate the 2007 EU Accession as the critical inflection point of regional divergence, they cannot inherently distinguish whether this stagnation was driven purely by the economic friction of the agricultural asset mismatch, or whether it was actively exacerbated by intentional political capture. To isolate the mechanism of local political clientelism, the empirical strategy employs a split-sample Difference-in-Differences design.

By partitioning the treated Plains counties based on their vulnerability to the political machine – proxied by their mean PSD vote share – this design tests whether the intensity of the post-2007 divergence scales directly with the strength of local clientelist networks. To do this, the pooled TWFE specification (like in Equation 3) is estimated separately for the "High-PSD" and "Low-PSD" subsamples to capture the distinct ATTs for each political cohort. Additionally, the dynamic event study framework (Equation 4) is applied to each subsample to visually confirm the parallel trends assumption within each specific split.

The core intuition of this split-sample approach relies on maintaining a consistent control group while varying the treatment intensity. Both the High-PSD and Low-PSD treated counties are benchmarked against the exact same 21 Mountain counties, which historically feature robust non-PSD competition and alternative economic safety nets.[26]

Finally, recent econometric advancements highlight that the standard TWFE estimator can be heavily biased in the presence of staggered treatment timing and heterogeneous treatment effects across cohorts (de Chaisemartin & D'Haultfœuille, 2023). Fortunately, because the identifying shock of this paper (EU Accession) occurs simultaneously for all treated units in the single year of 2007, this design inherently avoids severe negative weighting issues associated with staggered rollouts. Consequently, valid causal identification in this standard 2×2 framework requires only the parallel trends assumption, the plausibility of which will be checked with the help of the visualization of pre-trends. Standard errors remain clustered at the county level.

[26]The sample splitting is based on the median PSD vote share among all 42 counties. Thus, High-PSD sample pairs 13 High-PSD Plains with the 21 Mountains (N=816 county-years; N=782 when including time-varying controls and Rural Business Formation). The Low-PSD sample pairs the remaining 6 Low-PSD Plains with the same 21 Mountains (N=648 county-years; N=621 with full controls). Variables with missing historical data, such as Rural Natural Increase, result in marginally lower observation counts.

D. Synthetic Difference-in-Differences

Even though the split-sample TWFE design effectively helps to capture the political mechanism, slicing the dataset into smaller political sub-cohorts introduces distinct econometric challenges. Specifically, the High-PSD sample relies on comparing just 13 treated Plains counties against the 21 Mountain controls. In such small-N environments, standard TWFE estimators can be inefficient and highly sensitive to the parallel trends assumption, as equally weighting all 21 Mountain counties may incorporate noisy or poorly-matched counterfactuals. Furthermore, heavily parameterized models with time-varying controls risk small-sample overfitting.

To resolve these limitations and rigorously validate the clientelism mechanism, the final stage of the empirical strategy applies the Synthetic Difference-in-Differences estimator developed by Arkhangelsky et al. (2021) primarily to the High-PSD sample, alongside a parallel estimation on the Low-PSD sample to serve as a strict falsification test to check robustness. SDiD improves upon both standard DiD and traditional Synthetic Controls methods by simultaneously estimating unit weights and time weights to construct an optimal, uncaptured counterfactual. The estimator is defined as:

$$(\hat{\tau}^{SDiD}, \hat{\mu}, \hat{\alpha}, \hat{\delta}) = \arg\min_{\tau,\mu,\alpha,\delta} \sum_{c,t} (Y_{ct} - \mu - \alpha_c - \delta_t - \tau W_{ct})^2 \, \hat{\omega}_c \hat{\lambda}_t \tag{5}$$

where W_{ct} is the binary treatment indicator (taking the value of 1 for the treated Plains sub-cohort post-2007), $\hat{\tau}^{SDiD}$ represents the causal treatment effect of EU Accession on that specific cohort, and μ captures the grand mean (overall intercept). By estimating separately for the Low-PSD sample, the framework mathematically verifies whether the divergence changes in the absence of the political machine.

The strength of the SDiD estimator lies in its dual-weighting mechanism. First, it calculates unit weights ($\hat{\omega}_c$), which are assigned exclusively to the donor pool (the 21 Mountain counties).[27] Instead of treating all Mountain counties equally, the algorithm mathematically selects and reweights a specific combination of counties to synthesize a "perfect" control group whose pre-2007 trajectory precisely mimics that of the High-PSD Plains. Second, it calculates time weights ($\hat{\lambda}_t$), which are applied to the pre-treatment periods to heavily weight the specific pre-shock years that are most predictive of post-shock outcomes, effectively stabilizing the pre-trend matching.

Consequently, this dual-weighting optimization renders the inclusion of the socioeconomic control vector ($\mathbf{X}_{ct}$) unnecessary. Given that the synthesized control group is constructed to perfectly match the treated group's pre-treatment structural evolution, the weights inherently absorb the confounding localized trajectories that the covariates previously attempted to proxy.

[27] The High-PSD estimation uses 13 treated Plains and 21 Mountain donors over 24 years (N=816). the Low-PSD falsification uses 6 treated Plains and the same donors (N=648). Business Formation N is lower due to missing early-panel data. To satisfy SDiD's balanced panel requirement, a single missing 1997 Neamț observation was linearly interpolated.

By dropping the controls, the SDiD framework helps to eliminate the risk of post-treatment bias and small-sample overfitting.

Finally, unlike standard TWFE models that rely on clustered standard errors, statistical inference in this SDiD framework is conducted using the Jackknife method. By iteratively dropping single units and re-estimating the model, the Jackknife specifically accounts for the variance introduced by the estimation of the weights themselves, providing more conservative and reliable standard errors even in a restricted 34-county panel.

VII. Results

A. Geographic Baseline

Establishment of the structural baseline is based on the OLS estimates of the geographic penalty, which are found in Table 4 – this and other models estimated with no controls can be found in Appendix D. Coefficients for the binary Plains dummy represent average level differences versus the Mountains, interpreted in native units: percentage points (privatization, PSD vote, fiscal dependency), units per 100 arable hectares (tractors), and per 1,000 inhabitants (businesses, migration, natural increase).

The table confirms a severe institutional and physical capital penalty in the Plains. The geographic dummy predicts a privatization rate 3.38 percentage points lower than the Mountains ($p < 0.01$), validating the spatial disparity. This stalled reform coincides with a substantial structural deficit of -1.78 tractors per 100 arable hectares – a 68% reduction relative to the sample mean. Notably, the privatization rate itself does not predict tractor density, which suggests that the capital deficit is a direct geographic consequence of the asset mismatch (the hold-up problem), not merely a symptom of slow administrative reform.

Conversely, enterprise creation reveals a counterintuitive mechanism. While geography alone does not significantly suppress rural business formation, the privatization rate, surprisingly, does. Although the coefficient seems small (-0.022), the fact that the average privatization rate varies across the Plains and the Mountains by up to five percentage points – even more if taking individual counties – makes this a meaningful magnitude. Because the Plains experienced significantly slower privatization, this negative coefficient presents an empirical puzzle: the flatlands actually possessed a latent structural advantage for enterprise creation. This suggests a complex trade-off: even though physical capital destruction was driven directly by the geographic asset mismatch, the outright suppression of rural businesses cannot be explained by delayed reform, leaving an anomaly that geography fails to resolve.

Finally, the estimates establish the spatial foundations of both political capture and demographic immobility. The Plains dummy predicts a 7.06 percentage point increase in PSD vote share – a relative 20% increase over the mean – incubating the clientelist machine. Simultaneously, the positive external net migration coefficient (0.354) indicates comparatively lower emigration or even higher immigration to the Plains. This empirically supports the liquidity constraint hypothesis: “asset-rich but cash-poor” households lacked the liquid means for international exit.

TABLE 4: OLS REGRESSION RESULTS

Dependent Variable:	(1) **Privatization Rate**	(2) **Tractor Density**	(3) **Business Formation**	(4) **External Migration**	(5) **Rural Natural Increase**	(6) **Fiscal Dependency**	(7) **PSD Vote Share**
Plains Dummy	-3.377***	-1.783***	0.319	0.354**	-0.288	-0.452	7.061**
	(1.132)	(0.417)	(0.208)	(0.154)	(1.211)	(0.525)	(2.721)
Privatization Rate		-0.008	-0.022**	0.007	0.082	-0.064	-0.201
		(0.028)	(0.011)	(0.008)	(0.058)	(0.044)	(0.186)
Dependent Variable Mean	94.399	2.589	1.386	-0.180	-4.301	12.892	32.125
	(7.378)	(1.631)	(0.744)	(1.078)	(3.777)	(10.115)	(13.075)
Observations	760	760	720	760	759	760	160
Adjusted R^2	0.535	0.542	0.436	0.140	0.251	0.911	0.554
Adjusted Within R^2	0.336	0.502	0.360	0.086	0.244	0.085	0.458
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County
No. of Counties	40	40	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include Time Fixed Effects and control variables.

However, core indicators like rural natural increase and fiscal dependency yield insignificant geographic baseline effects. Geography laid the trap and initialized the lock in immobility, but realizing the ultimate demographic and fiscal stagnation required exogenous 2007 EU Accession shock and the active enforcement of political capture.

B. Asset Mismatch Channel

Having established the geographic baseline, it is useful to now turn to the dynamic macroeconomic shock caused by Romania joining the European Union. Figure 11 and Table 5 help evaluate whether integration into the European single market resolved Plains' asset mismatch or accelerated its collapse.

Figure 11 (left column) visualized the dynamic event study trajectories across three distinct institutional phases: restricted mobility (1996-2001), the visa-free intermediary regime (2002-2007), and post-EU Accession (2007 onwards). For modern developmental indicators – such as business formation, natural increase, and fiscal dependency – the Plains and Mountains exhibit reassuringly parallel pre-trends across both pre-shock phases. Tractor density exhibits a slight downward pre-trend from 1996 to 2001, reflecting the steady depreciation of the inherited socialist fleet without replacement. However, this trajectory stabilizes onto the zero-line during the 2002-2007 intermediary period. This confirms that on the eve of EU Accession, parallel trends held across all core outcomes.

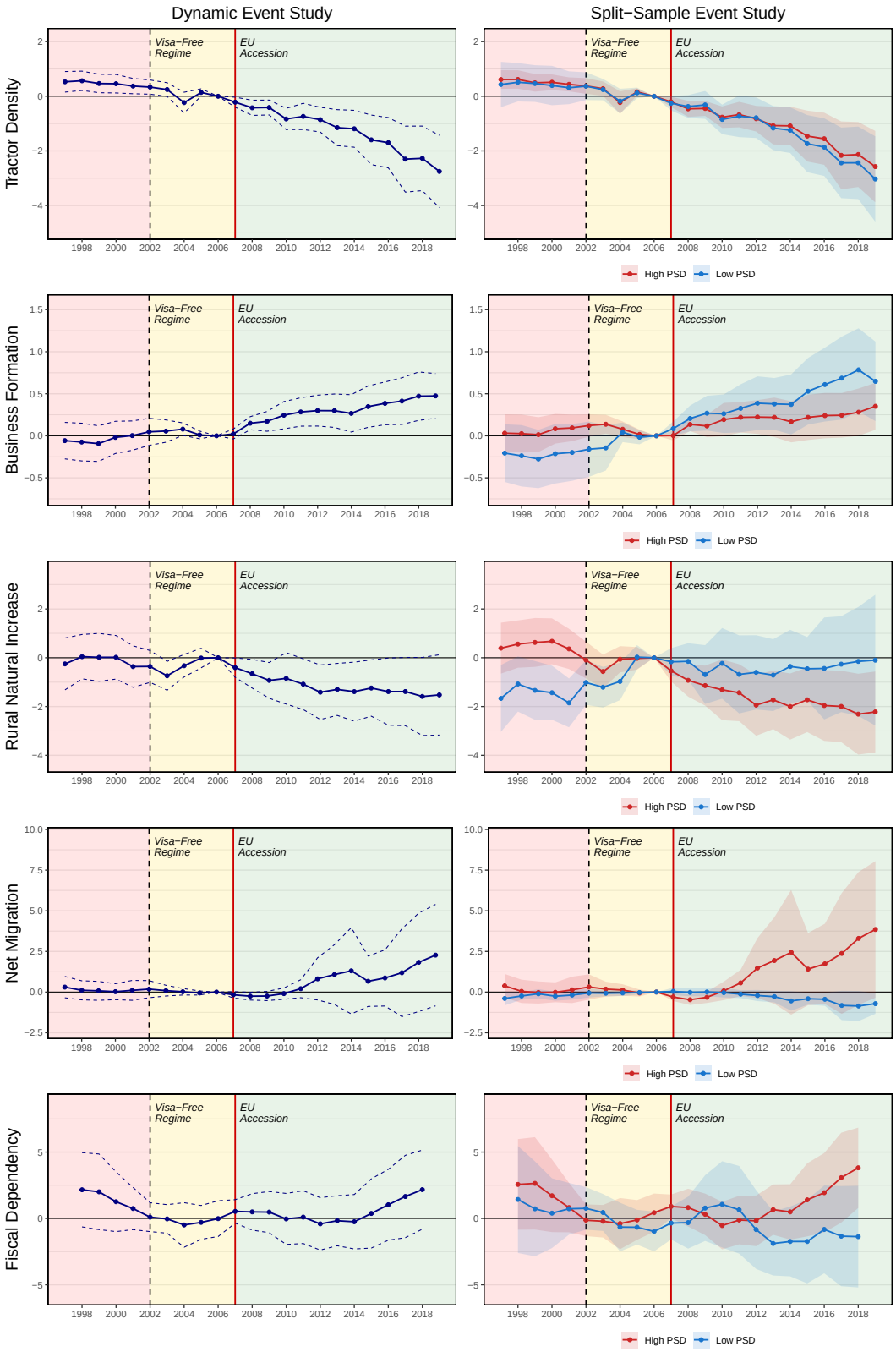


FIGURE 11: DYNAMIC & SPLIT-SAMPLE EVENT STUDIES

Notes: Left column: aggregate event study estimates (Equation 4). Right column: split-sample estimates partitioning the Plains into High- and Low-PSD cohorts. Solid lines and markers denote point estimates (β_k), with 95% confidence intervals shown as dashed lines (left) and shaded regions (right). To filter short-term volatility, Fiscal Dependency is smoothed via a 3-year centered moving average.

Once formal accession occurred, the trade shock acted as a significant catalyst for agricultural divergence. Table 5 (Column 1) confirms that exposing the Plains to open European competition influenced a persistent collapse in physical capital, dropping tractor density by an additional -1.556 units per 100 hectares ($p < 0.01$). The exogenous market shock exposed the latent vulnerability of the fragmented agricultural system, finalizing the destruction of the region's capital stock. Yet, interestingly, the aggregate Pooled DiD reveals a statistically significant increase in rural business formation (0.300). This complements the empirical paradox: if the core agrarian foundation of the Plains was collapsing, why was aggregate enterprise creation rising?

The dynamic post-treatment graphs for demographics and local institutions deepen this puzzle. While their pooled estimators lack statistical significance, their visual trajectories reveal distinct structural patterns shaped by broader macroeconomic headwinds. Rural natural increase experiences a clear downward post-shock trend, indicating a developing demographic decay. Concurrently, net migration trends upward – suggesting comparatively less emigration or more return migration to the Plains – though the wide confidence intervals reflect high spatial variance.

TABLE 5: POOLED DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Plains × Post-2007	-1.556***	0.300**	0.666	-0.976	0.007
	(0.331)	(0.115)	(0.537)	(0.598)	(0.730)
Dependent Variable Mean	2.898	1.546	-0.005	-4.385	12.785
	(2.079)	(0.851)	(1.846)	(3.751)	(8.773)
Observations	920	920	920	919	920
Adjusted R^2	0.822	0.921	0.337	0.899	0.876
Adjusted Within R^2	0.185	0.155	0.038	0.073	0.056
Controls	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects, as well as control variables.

Finally, fiscal dependency traces a distinct U-shaped curve. The initially high dependency falls steadily during the early 2000s as Romania implemented the strict macroeconomic stabilization and austerity policies required to qualify for EU membership. Then it remains flat immediately following accession – likely as the 2008 global financial crisis suppressed state revenues uniformly – but begins trending upward after 2012. As the Eurozone Sovereign Debt Crisis severely impacted primary migrant destinations (Italy and Spain), straining external

remittance channels, and as EU integration expanded the central resources available for local patronage, reliance on state transfers in the structurally weak Plains swelled once again.

Ultimately, these noisy aggregate demographics, the fiscal U-curve, and the paradoxical rise in businesses indicate that evaluating the Plains as a monolithic geographic block is analytically incomplete. The aggregate treatment effect hides important institutional heterogeneity. To understand how the physical capital collapse was actively instrumentalized into an immobile, dependent "Fiscal Zombie" equilibrium, the analysis moves beyond geography and isolates the local political machine.

C. Political Capture Channel

Figure 11 (right column) and Table 6 attempt to resolve the aforementioned puzzle by partitioning the Plains into High-PSD and Low-PSD sub-cohorts, revealing that the aggregate treatment effect hides two, effectively, different institutional trajectories.

The split-sample event studies demonstrate that the destruction of the agrarian base was purely geographic, but the subsequent recovery was rather political. Looking at tractor density in Figure 11, the trajectories of both cohorts are virtually identical. The slight pre-2002 aggregate parallel trend violation is shown to be driven entirely by the High-PSD cohort, which possessed a slightly larger initial stock of aging machinery. Post-2007, both cohorts suffer a nearly identical, statistically significant collapse in their pooled tractor density coefficients. Politics could not save indivisible socialist infrastructure from the geographic asset mismatch.

However, a divergence emerges in enterprise creation. While both cohorts exhibit parallel pre-trends, the pooled estimator for post-shock business formation surges significantly in the Low-PSD cohort. Conversely, the High-PSD trajectory is more muted, remaining lower and frequently touching the zero bound. This helps to solve the aggregate puzzle: when the agricultural base collapsed, politically competitive counties successfully transitioned toward, for example, non-farm commercial enterprises. In captured counties, the political machine stifled this commercial exit strategy, trying to preserve a monopoly on local economic opportunity.

By blocking enterprise, the political apparatus plausibly contributed to engineering a demographic trap. In the Low-PSD cohort, rural natural increase and net migration remain remarkably flat and stable post-shock (notably, Low-PSD natural increase violates pre-trends by converging diagonally before 2007, but stabilizes completely along the zero line post-accession). In contrast, the High-PSD pooled coefficient for natural increase captures a substantial and statistically significant demographic decay. At the same time, High-PSD net migration experiences an upward jump in magnitude, almost mirroring the aggregate event study, though its pooled coefficient remains statistically insignificant due to wide spatial variance.[28]

[28] While this localized upward migration trend strongly supports the liquidity constraint hypothesis, it may also be partially inflated by the post-2007 influx of Moldovan citizens acquiring Romanian passports and settling in eastern border counties. Nevertheless, the structural immobility of the native population remains clear.

TABLE 6: SPLIT-SAMPLE DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	**Tractor Density**		**Business Formation**		**External Migration**		**Rural Natural Increase**		**Fiscal Dependency**	
PSD Level:	**High** (1)	**Low** (2)	**High** (3)	**Low** (4)	**High** (5)	**Low** (6)	**High** (7)	**Low** (8)	**High** (9)	**Low** (10)
Plains × Post-2007	-1.497***	-1.585***	0.140	0.564***	1.282	-0.208	-1.801***	0.642	0.552	-1.214
	(0.293)	(0.485)	(0.126)	(0.136)	(0.788)	(0.155)	(0.616)	(0.682)	(0.823)	(0.917)
Observations	782	621	782	621	782	621	781	620	782	621
Adjusted R^2	0.825	0.783	0.912	0.933	0.365	0.315	0.918	0.895	0.877	0.881
Adjusted Within R^2	0.188	0.093	0.092	0.315	0.077	0.010	0.168	0.127	0.064	0.041
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	34	27	34	27	34	27	34	27	34	27

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects, as well as control variables.

The ultimate objective of this demographic trap is visible in the fiscal dependency trajectories. The Low-PSD cohort experiences relative fiscal stability, hovering around the zero line from 1997 through 2018. The delayed aggregate U-curve identified in the previous subsection was plausibly fueled by the High-PSD cohort. As the captured population became increasingly locked-in, High-PSD dependency started to increase after 2012. While the overall pooled coefficient remains statistically insignificant – diluted by the flatline of the 2008 crisis – the dynamic point estimates become significant by 2018.[29]

Hence, the split-sample visually dissects the anatomy of the "Fiscal Zombie" equilibrium. Geography destroyed the agricultural base, but it was the PSD machine that utilized this collapse – suppressing enterprise to manufacture a captive, immobile, and state-dependent electorate. However, because standard TWFE estimators struggle to achieve strict statistical significance across all pooled variables in a restricted, small-N split-sample, the final stage of analysis applies the SDiD estimator to High-PSD Plains in order to definitively eliminate pre-trend noise and rigorously quantify the causal magnitude of this political trap.

D. Quantifying the Fiscal Zombie

Figure 12 plots the observed trajectories against their optimized synthetic counterfactuals not only for the main High-PSD specification, but also for Low-PSD as a falsification test. Formal estimates for the High-PSD cohort are presented in Table 7. The coefficients for the falsification test can be found in Appendix C.

[29] Methodologically, the exceptionally high aggregate R^2 for fiscal dependency across all specifications (including the baseline OLS) reflects the deeply persistent, structural nature of local state revenues. The strictly bounded Adjusted Within-R^2 confirms that this high aggregate fit successfully absorbs baseline heterogeneity rather than indicating harmful multicollinearity among the covariates.

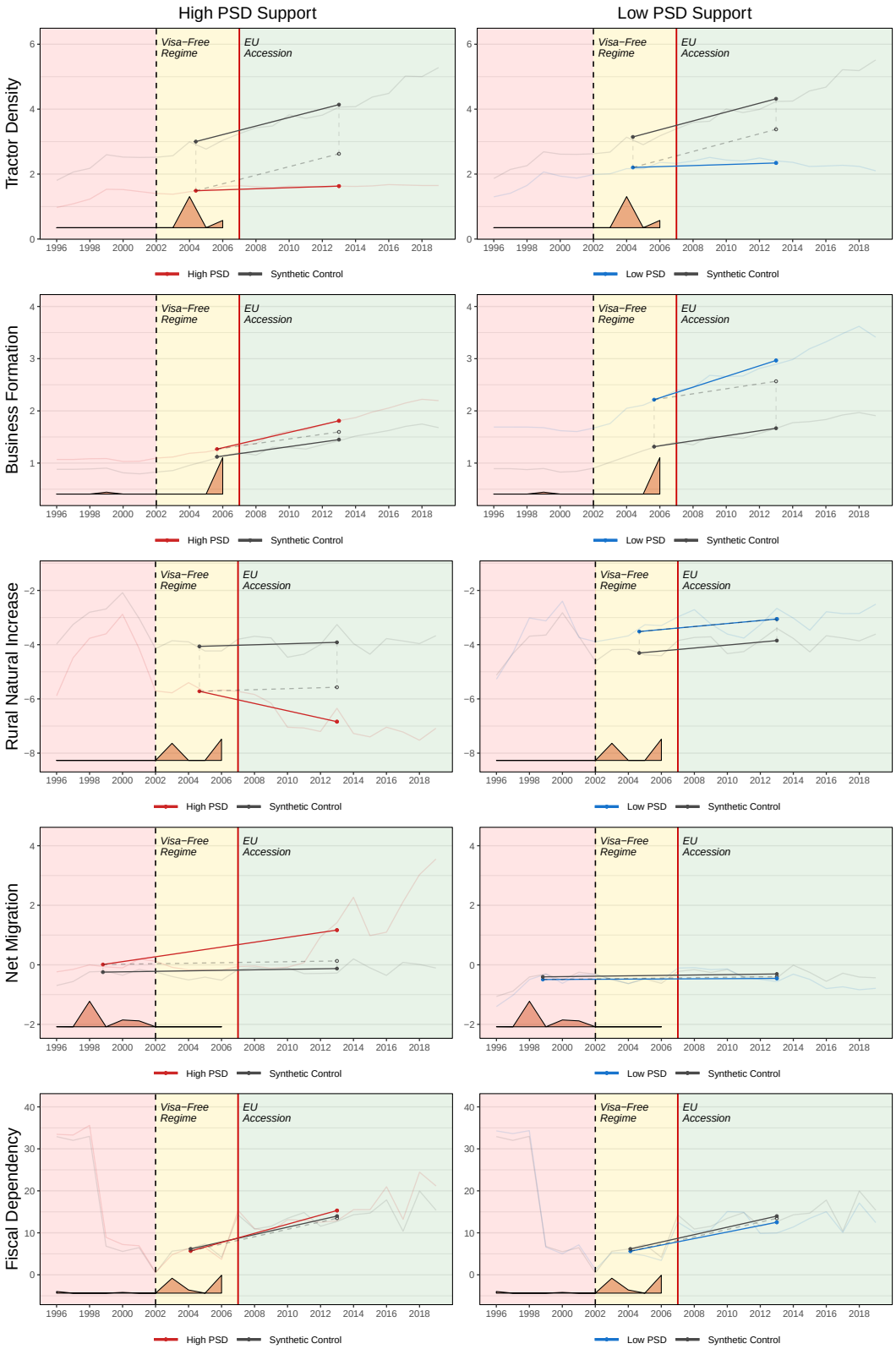


FIGURE 12: SYNTHETIC DID TRAJECTORIES

Notes: These figures plot the observed actual trajectories of the treated Plains sub-cohorts against their respective Synthetic Controls constructed via the SDiD estimator (Equation 5). The left column evaluates the High-PSD sample, while the right column evaluates the Low-PSD sample as a falsification test.

The SDiD framework confirms that the destruction of physical capital was a universal geographic penalty, but the suppression of enterprise was a deliberate political choice. Post-2007, the High-PSD tractor trajectory completely flatlines, resulting in a severe penalty (-0.994, $p < 0.01$), although smaller in magnitude than previous models, as the Synthetic Plains counterfactual successfully begins re-mechanizing. While the Low-PSD falsification cohort also loses tractors, it stabilizes at a higher threshold.

Political capture mechanism becomes evident when observing rural business formation. The Low-PSD falsification trajectory shows that the Plains possessed a significant latent commercial advantage, surging well above its synthetic counterfactual post-shock. Conversely, the High-PSD cohort's business formation is lower and quite similar to that of the Synthetic Plains. While it manages a statistically significant positive effect against its synthetic baseline (0.213), its slope and magnitude are rather suppressed relative to the region's true non-farm potential. The local political apparatus allowed just enough enterprise to function, but essentially restrained structural commercialization to maintain its monopoly on economic leverage.

TABLE 7: SYNTHETIC DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
SDiD Estimate ($\hat{\tau}^{SDiD}$)	-0.994***	0.213**	1.037	-1.270**	1.829**
	(0.220)	(0.094)	(0.743)	(0.513)	(0.834)
Observations	816	782	816	816	816
Treated Counties	13	13	13	13	13
Donor Pool (Mountains)	21	21	21	21	21
Time Weights ($\hat{\lambda}_t$)	Estimated	Estimated	Estimated	Estimated	Estimated
Unit Weights ($\hat{\omega}_c$)	Estimated	Estimated	Estimated	Estimated	Estimated
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Standard Errors	Jackknife	Jackknife	Jackknife	Jackknife	Jackknife

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Jackknife standard errors are reported in parentheses. The model matches High-PSD Plains to a Synthetic Mountain Baseline.

By stifling this commercial exit, the machine finalized a demographic trap. Even though the Low-PSD cohort tracks its synthetic demographic counterfactual, the High-PSD cohort structurally collapses. Rural natural increase breaks from the synthetic baseline post-accession, falling into a worsening decay (-1.270 – a fall equivalent to approximately 30% of the total sample mean). Simultaneously, net migration experiences a visible post-2010 upward jump against its synthetic control, indicating a reduction in out-migration and increase in immigration. Together, these two dynamics describe a demographic reality defined by captured and immobile population that is rapidly aging and no longer reproducing.

The key aim of this trapped, dying demographic base is realized in the fiscal trajectories. While the Low-PSD cohort becomes relatively more fiscally independent than its synthetic

counterfactual, the High-PSD cohort seems to diverge upward from its synthetic baseline after 2014 the same way it did in the previous subsection. With statistical noise mathematically eliminated, the SDiD estimator confirms a considerable and highly significant increase in local fiscal dependency by 1.829 percentage point. Against a baseline sample mean of 12.89%, this, thus, represents a visible structural shift toward state reliance by an increase of about 14%.

VIII. Robustness Checks

In order to verify that the final equilibrium is not an artifact of sample selection or model specification, it is important to establish essential robustness checks in addition to the falsification test discussed previously. Hence, this section subjects the core empirical mechanisms to four rigorous sensitivity tests. The detailed results of these tests can be found in Appendix C.

A. Extreme TRI Split

First, a common threat to spatial DiD designs is the potentially arbitrary nature of geographic boundaries. The binary "Plains against Mountains" classification risks mischaracterizing counties situated in transitional, hilly regions. If these middle-tier counties are misclassified, they could artificially drive or suppress the aggregate geographic penalty observed in the baseline estimates.

To resolve this boundary ambiguity, we utilize the TRI to execute an extreme subsample analysis. We strictly drop the middle 50% of the aggregate sample, isolating and re-estimating the baseline DiD model exclusively on the flattest 25% (Extreme Plains) against the most rugged 25% (Extreme Mountains).

The results in Appendix C validate the geographic origins of the asset mismatch. Eliminating transitional geographies actively magnifies the baseline physical capital penalty. The Extreme Plains exhibit a post-2007 tractor deficit of -2.368 ($p < 0.01$) with full controls, a substantial intensification compared to the baseline aggregate penalty of -1.556. This confirms that the agricultural collapse is plausibly a structural geographic phenomenon rather than a fuzzy boundary artifact. Interestingly, rural business formation in these flattest counties also increases significantly (0.501). Given our previous finding that enterprise creation was fueled almost entirely by politically competitive (Low-PSD) counties, this amplified coefficient indicates that the least rugged plains actually might possess the greatest latent potential for non-farm commercialization. Simultaneously, demographic and fiscal outcomes remain statistically indistinguishable from zero across this extreme spatial divide. This supports the foundational premise of the paper: while extreme terrain influenced the initial capital mismatch and the destruction of heavy agriculture, geography alone cannot explain the subsequent demographic trap.

B. Extreme PSD Split

Even though the main split-sample analysis established the presence of political capture, categorizing all Plains counties as simply "High" or "Low" PSD risks diluting the true magnitude of the mechanism. Counties hovering near the median political threshold might blur the insti-

tutional divergence. If the "Fiscal Zombie" theory holds, the suppression of enterprise and the resulting demographic collapse should act as a dose-response mechanism: scaling in severity alongside the intensity of clientelist capture.

To test this, we isolate the extremes of the political spectrum. We drop the middle 50% of the Plains counties, running the split-sample model strictly on the top 25% (Severely Captured) against the bottom 25% (Uncaptured) against the standard Mountains control group.

The extreme dose-response estimates in Appendix C validate the anatomy of the clientelist trap. The destruction of physical capital remains universal, yielding slightly more severe or unchanged, but still statistically significant tractor deficits in both heavily captured (-1.563) and uncaptured (-1.500) extremes. However, the economic recovery is rather bifurcated. Rural business formation increases significantly (0.520) in the uncaptured plains but is indistinguishable from zero in the captured plains, confirming the original findings. Consequently, denied both heavy agriculture and independent enterprise, the captured plains experience an intensified collapse in rural natural increase which is even greater than in the main findings (-2.322 against -1.801), whereas the uncaptured plains remain more stable. By isolating the extreme tails of political control, stable or even more severe coefficients help to prove that clientelist networks plausibly gatekept EU-era economic adaptation to engineer a captive demographic base.

C. Truncated Timeline

Nevertheless, the fact that the first decade of Romania's post-communist transition was notoriously volatile needs to be taken into account, too. The 1990s were characterized by severe macroeconomic instability, hyperinflation, and chaotic deindustrialization (Gallagher, 2005; Pop-Eleches, 2007). Thus, the observed divergence might simply capture a delayed hangover from this turbulent decade rather than a true structural response to the 2007 EU Accession.

In order to definitively separate the EU integration era from early-transition noise, we permanently truncate the dataset by dropping all observations prior to 2000. The baseline DiD model is then re-estimated exclusively on the restricted 2000-2019 window.

As reported in Appendix C, dropping the hyper-volatile 1990s leaves the core mechanisms statistically undisturbed. The tractor density penalty remains highly significant and changed only marginally (-1.460, $p < 0.01$), confirming that the agricultural divergence was causally affected by open market exposure rather than 1990s events. Furthermore, the structural adaptation toward rural enterprise survives (0.273), validating the theory that independent entrepreneurship was a recent adaptation incentivized by EU Accession and plausibly its structural funds. Core demographic and fiscal parameters maintain their baseline insignificance. Overall, the restricted timeline confirms that the capital collapse and subsequent institutional divergence are reliable functions of the post-accession era.

D. Leave-One-Out Analysis

Finally, in a geographically restricted panel (N=40 counties), baseline estimates risk being artificially driven by a single high-leverage outlier. This is a particular concern for the exter-

nal migration parameter – it is plausible that single eastern border county could be skewing the aggregate migration variance due to the localized post-2007 influx of Moldovan citizens acquiring Romanian passports.

To test for outlier sensitivity, we systematically drop one county at a time from the panel and iteratively re-estimate the baseline model, generating 40 distinct point estimates for each dependent variable which are visualized in Figures B5-B9 of Appendix C.

The Leave-One-Out (LOO) plots demonstrate that the baseline findings are stable and systemic. For physical capital destruction, the tractor density point estimate hovers reliably around -1.6, with the 95% confidence intervals strictly excluding the zero threshold regardless of which county is excluded. The positive surge in aggregate business formation is equally stable. Rural natural increase and fiscal dependency parameters remain indistinguishable from zero no matter which county is excluded from the sample.

Furthermore, the LOO analysis helps to address the migration anomaly. Iteratively dropping individual eastern border counties (such as Vaslui, Iași, or Galați) does not alter the wide confidence intervals or the statistical insignificance of the net migration rate. This suggests that the high variance in migration is not merely a statistical artifact of a single border anomaly. Ultimately, the agricultural stagnation and subsequent economic bifurcation in the Plains are robust macro-level trends, plausibly independent of any single local administration.

IX. Conclusion

Romania's transition from 1950s forced collectivization to European Union integration left behind a noticeably unequal and structurally fractured rural landscape. This thesis, in particular, investigated the long-term consequences of the 1991 land restitution, identifying how geographic constraints and local political capture interacted following the exogenous macroeconomic shock of the 2007 EU Accession. By utilizing Baseline and Synthetic DiD frameworks, the analysis tackled a particular puzzle: why the agricultural plains collapsed into demographic decay and fiscal dependency, while the rugged mountainous regions structurally adapted.

The empirical findings demonstrate that the destruction of rural physical capital was a substantial and persistent geographic penalty. The egalitarian fragmentation of land in 1991 created a significant asset mismatch in the plains, significantly reducing the productive value of indivisible socialist agricultural machinery and resulting in an approximately 68% baseline collapse in regional tractor density relative to the mean. However, the subsequent economic trajectory of these regions was rather determined by local political institutions. Even though politically competitive plains successfully transitioned toward non-farm rural enterprises, clientelist networks in captured counties were associated with substantially weaker commercial adaptation, with business formation approximately half that of their less-captured counterparts. Consistent with a "Fiscal Zombie" equilibrium, counties with deeper political capture showed patterns of gatekept EU structural funds and suppressed private enterprise, as well as, on average, a 14% relative increase in local state reliance. The results are consistent with an immobile, aging

population becoming increasingly dependent on state transfers, supporting the hypothesis of a captive electorate. Consequently, this paper contributes to the broader political economy literature by showing that while geography influences the anatomy of an initial economic shock, political capture affects the possibility of long-term structural survival.

Despite the robust findings, the analysis faces limitations that offer fertile ground for future research. Primarily, the demographic mechanism – specifically the external migration parameter – apart from being a total and not a rural variable, is potentially affected by spatial noise due to the post-2007 influx of Moldovan citizens acquiring Romanians passports and settling in eastern border counties. Future empirical work could develop granular filtering mechanisms to separate native rural out-migration from this cross-border administrative artifact. A further limitation concerns the use of PSD vote share as a proxy for political capture, which remains potentially endogenous to the same economic stagnation it is meant to explain. Thus, future research could address this by instrumenting PSD dominance with predetermined measures of communist organizational density – such as county-level Party membership rates or *Securitate* officer presence in the 1950s-60s – which plausibly predict clientelist network persistence through institutional continuity while remaining orthogonal to post-accession outcomes. Moreover, exploring the broader regional context through comparative studies with other post-communist states – such as the Republic of Moldova or Bulgaria – could determine whether the clientelist demographic trap is a uniquely Romanian institutional legacy or a systemic feature of delayed agrarian transitions.

Finally, to completely resolve the institutional puzzle of Romanian agriculture, future research could examine the historical prequel to this study: the violent collectivization process itself. As data and historical context provided in Section III showed, there is a highly promising avenue for future doctoral research that could involve leveraging newly accessible archival data on peasant arrests, resistance, and forced collectivization quotas. Quantifying the initial destruction of human and economic capital during the 1950s would aid in connecting the historical dots, mapping the full, unfortunate trajectory of the Romanian village from its initial communist subjugation to its modern post-communist bifurcation.

References

Acemoglu, D., Johnson, S. & Robinson, J. A. (2001) The Colonial Origins of Comparative Development: An Empirical Investigation. Cambridge, *Mass: National Bureau of Economic Research*.

Adamopoulos, T. & Restuccia, D. (2020) Land Reform and Productivity: A Quantitative Analysis with Micro Data. *American economic journal. Macroeconomics. 12* (3), 1–39.

Angelucci, M. (2015) Migration and Financial Constraints: Evidence from Mexico. *The review of economics and statistics. 97* (1), 224–228.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W. & Wager, S. (2021) Synthetic Difference-in-Differences. *American Economic Review, 111* (12), 4088–4118.

Banerjee, A. & Lakshmi Iyer (2005) History, Institutions, and Economic Performance: The Legacy of Colonial Land Tenure Systems in India. *The American economic review. 95* (4), 1190–1213.

Besley, T. (1995) Property Rights and Investment Incentives: Theory and Evidence from Ghana. *The Journal of political economy. 103* (5), 903–937.

Bleakley, H. & Ferrie, J. (2014) Land Openings on the Georgia Frontier and the Coase Theorem in the Short- and Long-Run. NBER Working Paper No. 19946. *Cambridge, MA: National Bureau of Economic Research*.

Borșa, S. (2013). The Sovietization of Romania. Case-study: The Collectivization of Agriculture. *Academic Journal of Interdisciplinary Studies, 2* (8), pp. 298-303.

Bramall, C. (2004) Chinese Land Reform in Long-Run Perspective and in the Wider East Asian Context. *Journal of agrarian change. 4* (1–2), 107–141.

Bryan, G., Chowdhury, S. & Mobarak, A. M. (2014) Under-investment in a Profitable Technology: The Case of Seasonal Migration in Bangladesh. *NBER Working Paper Series*. 20172.

Callaway, B. & Sant'Anna, P. H. C. (2021) Difference-in-Differences with multiple time periods. *Journal of econometrics. 225* (2), 200–230.

Carter, M. R. & Barrett, C. B. (2006) The economics of poverty traps and persistent poverty: An asset-based approach. *The Journal of development studies. 42* (2), 178–199.

Cartwright, A. L. (2017) The Return of the Peasant: Land Reform in Post-Communist Romania. *1st edition. United Kingdom: Routledge*.

De Chaisemartin, C. and d'Haultfoeuille, X. (2023) Credible Answers to Hard Questions: Differences-in-Differences for Natural Experiments. *New Jersey: Princeton University Press*.

Deininger, K. & Songqing Jin (2006) Tenure security and land-related investment: Evidence from Ethiopia. *European Economic Review. 50* (5), 1245-1277.

Deletant, D. (2019) Romania under Communism: Paradox and Degeneration. *London: Routledge*.

Dell, M. (2010) The Persistent Effects of Peru's Mining Mita. *Econometrica. 78* (6), 1863–1903.

De Soto, H. (2001) Dead Capital and the Poor. *SAIS Review. 21* (1), 13–44.

Dobrincu, D. (2004) Dinamica represiunii comuniste împotriva țărănimii din România (1951–1959). *Arhiva Istorică a României, serie nouă. 1* (2), 253–267.

Dustmann, C. & Görlach, J.-S. (2016) The Economics of Temporary Migrations. *Journal of economic literature. 54* (1), 98–136.

Dutronc-Postel, P. & Zhuang, M. (2024) Cohorts, competition and construction: economic performance, land expropriation and bureaucrat promotion in China. Draft version. *Institut des politiques publiques & Stockholm Institute of Transition Economics*.

Ferguson, J. & Kim, O. (2023). Reassessing China's Rural Reforms: The View from Outer Space. *UC Berkeley*.

Foster, A. D. & Rosenzweig, M. R. (2022) Are There Too Many Farms in the World? Labor Market Transaction Costs, Machine Capacities, and Optimal Farm Size. *The Journal of political economy. 130* (3), 636–680.

Frydman, R., Gray, C., Hessel, M. & Rapaczynski, A. (1999) When Does Privatization Work? The Impact of Private Ownership on Corporate Performance in the Transition Economies. *Quarterly Journal of Economics, 114* (4), 1153–1191.

Gallagher, T. (2005) Theft of a nation: Romania since communism. *London: C. Hurst*.

Gorton, M. & Davidova, S. (2004) Farm productivity and efficiency in the CEE applicant countries: a synthesis of results. *Agricultural economics. 30* (1), 1–16.

Gow, H. R. & Swinnen, J. F. M. (1998) Up- and downstream restructuring, foreign direct investment, and hold-up problems in agricultural transition. *European review of agricultural economics. 25* (3), 331–350.

Grosfeld, I. & Zhuravskaya, E. (2015) Cultural vs. economic legacies of empires: Evidence from the partition of Poland. *Journal of Comparative Economics. 43* (1), 55–75.

Grossman, S. J. & Hart, O. D. (1986) The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration. *The Journal of political economy. 94* (4), 691–719.

Grzymala-Busse, A. (2008) Beyond Clientelism: Incumbent State Capture and State Formation. *Comparative political studies. 41* (4–5), 638–673.

Gusti, D. (ed.) (1938) Enciclopedia României. Volumul III: Economia Națională. București, *Imprimeria Națională*.

Hartvigsen, M. (2014) Land reform and land fragmentation in Central and Eastern Europe. *Land Use Policy. 36*, 330-341.

Heldring, L., Robinson, J. A. & Vollmer, S. (2022) The Economic Effects of the English Parliamentary Enclosures. Cambridge, *Mass: National Bureau of Economic Research*.

Heldring, L., Robinson, J. A. & Vollmer, S. (2015) The Long-Run Impact of the Dissolution of the English Monasteries. Cambridge, *Mass: National Bureau of Economic Research*.

Hirschman, A. O. (1970) Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States. *Cambridge, MA: Harvard University Press*.

Hornbeck, R. & Naidu, S. (2014) When the Levee Breaks: Black Migration and Economic Development in the American South. *The American economic review. 104* (3), 963–990.

Institutul Central de Statistică (1938) Recensământul general al populaţiei României din 29 decembrie 1930. *Monitorul Oficial, Imprimeria Naţională, volume 2*.

Ioniță, N. & Mărcuță, L. (2023) Studiu privind evoluția producției de cereale a României în perioada 2018-2022 și influența acesteia asupra rentabilității exploatațiilor agricole. *An. INCDA Fundulea, 91* (1), p. 127-139.

Iordachi, C. & Dobrincu, D. (2009) Transforming peasants, property and power: the collectivization of agriculture in Romania, 1949-1962. 1st ed. *Budapest; Central European University Press*.

Ivănescu, S.D. (2015). Implicarea Securității în procesul de colectivizare a agriculturii din România. *Anuarul Institutului de Istorie A.D. Xenopol - Iași, 52*, p. 383-396.

Iyer, L. & Mani, A. (2012) Traveling Agents: Political Change and Bureaucratic Turnover in India. *The review of economics and statistics. 94* (3), 723–739.

Klein, B., Crawford, R. G. & Alchian, A. A. (1978) Vertical Integration, Appropriable Rents, and the Competitive Contracting Process. *The Journal of law & economics. 21* (2), 297–326.

Kligman, G. & Verdery, K. (2011) Peasants under siege: the collectivization of Romanian agriculture, 1949-1962. *Princeton, N.J: Princeton University Press*.

Manuelli, R. E. & Seshadri, A. (2014) Frictionless Technology Diffusion: The Case of Tractors. *The American economic review. 104* (4), 1368–1391.

Mares, I. & Young, L. E. (2019) Conditionality & coercion: electoral clientelism in Eastern Europe. *Oxford [etc: Oxford University Press*.

Markevich, A. & Zhuravskaya, E. (2018) The Economic Effects of the Abolition of Serfdom: Evidence from the Russian Empire. *The American economic review. 108* (4–5), 1074–1117.

Martínez, L. R. (2022) How Much Should We Trust the Dictator's GDP Growth Estimates? *The Journal of political economy. 130* (10), 2731–2769.

Mathijs, E. & Swinnen, J. F. M. (1998) The Economics of Agricultural Decollectivization in East Central Europe and the Former Soviet Union. *Economic Development and Cultural Change, 47* (1), 1–26.

Nunn, N. (2008) The Long-Term Effects of Africa's Slave Trades. Cambridge, *Mass: National Bureau of Economic Research*.

O'Dwyer, C. (2006) Runaway state-building: patronage politics and democratic development. 1st ed. Baltimore, *Md: Johns Hopkins University Press*.

Pop-Eleches, G. (2007) Historical Legacies and Post-Communist Regime Change. *The Journal of politics. 69* (4), 908–926.

Riley, S. J., DeGloria, S. D. & Elliot, R. (1999) A terrain ruggedness index that quantifies topographic heterogeneity. *Intermountain Journal of Sciences, 5* (1–4), 23–27.

Rizov, M., Pokrivcak, J. & Ciaian, P. (2013) CAP Subsidies and Productivity of the EU Farms. *Journal of agricultural economics. 64* (3), 537–557.

Roberts, H. L. (1951) Rumania: Political Problems of an Agrarian State. *New Haven: Yale University Press*.

Robinson, J. A. & Verdier, T. (2013) The Political Economy of Clientelism. *The Scandinavian journal of economics. 115* (2), 260–291.

Sabates-Wheeler, R. (2002). Consolidation initiatives after land reform: responses to multiple dimensions of land fragmentation in Eastern European agriculture. *Journal of International Development, 14* (7), 1005-1018.

Sandu, D. (2005) Emerging Transnational Migration from Romanian Villages. *Current sociology. 53* (4), 555–582.

Serviciul Român de Informații (1994) Cartea Albă a Securității (1948-1958), *București, volume 2*, p. 423-426.

Suret-Canale, J. (1956) Le recensement du 21 février 1956 en Roumanie. *L'information géographique, volume 20* (4), 1956, p. 146.

Stokes, S. C., Dunning, T., Nazareno, M. & Brusco, V. (2013) Brokers, Voters, and Clientelism: The Puzzle of Distributive Politics. *Cambridge: Cambridge University Press*.

Swinnen, J.F.M. & Rozelle, S. (2006) From Marx and Mao to the market: The economics and politics of agricultural transition. *Oxford: Oxford University Press*.

Urșanu, E.-A., Grigorescu, I., Dumitrică, C., Kucsicsa, G., Mitrică, B., Roznovietchi, I., Dumitrașcu, M. & Ciubuc, C. (2024) Long-term changes of agricultural land over the last century in Romania. The showcase of Romanian plain. *Anthropocene. 48*, 100449.

Verdery, K. (2003) The vanishing hectare: property and value in postsocialist Transylvania. Ithaca, *N.Y. [etc: Cornell U.P]*.

Wegren, S. K. (2005) The moral economy reconsidered: Russia's search for agrarian capitalism. 1st ed.

Williamson, O. E. (1987) The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting. *New York: The Free press*.

Zimmerman, F. J. & Carter, M. R. (2003) Asset smoothing, consumption smoothing and the reproduction of inequality under risk and subsistence constraints. *Journal of Development Economics, 71* (2), 233–260.

Appendix

A. Variable Construction and Sources

TABLE A1: CLASSIFICATION OF COUNTIES BY TERRAIN RUGGEDNESS INDEX (TRI)

Plains (TRI < Median)		Mountains (TRI > Median)	
County	TRI Score	County	TRI Score
Brăila	3.2	Mehedinți	26.5
Ialomița	3.6	Vrancea	29.6
Călărași	3.9	Sălaj	31.4
Ilfov	4.6	Prahova	31.9
Giurgiu	4.9	Covasna	34.5
Teleorman	5.0	Mureș	34.8
Bucharest	5.1	Bacău	37.6
Olt	7.1	Cluj	37.6
Dolj	8.4	Argeș	38.6
Tulcea	9.2	Brașov	39.3
Constanța	9.5	Sibiu	41.5
Timiș	10.3	Harghita	41.5
Galați	11.3	Neamț	41.6
Satu Mare	13.5	Gorj	42.7
Botoșani	15.6	Suceava	43.8
Iași	17.5	Caraș-Severin	48.4
Dâmbovița	18.5	Alba	48.5
Arad	19.6	Vâlcea	49.5
Vaslui	21.2	Bistrița-Năsăud	50.3
Bihor	25.7	Maramureș	50.3
Buzău	26.3	Hunedoara	56.5

Notes: The sample is split at the median TRI value.

TABLE A2: DESCRIPTIVE STATISTICS

	Obs.	Mean	Std. Dev	Min	Median	Max
Panel A: Full Sample						
Core Variables						
Privatization Rate (1996-1999)	166	86.14	10.43	58.64	87.92	99.25
Privatization Rate (1996-2014)	796	93.85	7.86	58.64	97.29	99.78
Tractor Density	1,004	3.92	16.31	0.46	2.18	483.33
Modern Variables						
Rural Business Formation	943	1.54	0.85	0.33	1.33	4.79
Net Migration Rate	1,008	-0.003	1.81	-3.25	-0.22	23.01
Rural Natural Increase	983	-4.34	3.74	-14.45	-4.26	5.86
Fiscal Dependency Ratio	1,008	13.40	9.71	0.04	11.30	51.50
PSD Vote Share	210	34.97	11.28	6.40	36.80	50.40
Control Variables						
Modernized Roads Ratio	1,008	59.60	13.07	33.45	58.72	100.00
High School Grads Ratio	1,008	10.54	2.85	3.59	10.27	20.57
Hospital Beds Index	966	5.91	1.68	2.62	5.62	11.11
Social Aid Index	1,006	32.19	30.25	0.48	23.32	237.13

	Plains (TRI < Median)			Mountains (TRI > Median)		
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev
Panel B: Comparison by TRI Score						
Core Variables						
Privatization Rate (1996-1999)	82	79.6	9.59	84	92.4	4.83
Privatization Rate (1996-2014)	397	91.0	6.54	399	96.6	2.48
Tractor Density	500	4.39	12.2	504	3.86	1.71
Modern Variables						
Rural Business Formation	460	1.78	0.82	483	1.31	0.52
Net Migration Rate	504	0.28	1.30	504	-0.29	0.38
Rural Natural Increase	480	-4.93	3.66	503	-3.78	3.35
Fiscal Dependency Ratio	504	13.30	2.43	504	13.4	2.20
PSD Vote Share	105	39.10	9.51	105	30.90	11.60
Control Variables						
Modernized Roads Ratio	504	62.40	13.50	504	56.70	8.02
High School Grads Ratio	504	9.95	1.40	504	11.10	1.02
Hospital Beds Index	483	5.68	1.66	483	6.15	1.39
Social Aid Index	502	25.60	5.17	504	38.70	13.40

Notes: Panel A reports summary statistics for the full sample (1996–2016). Panel B divides the sample by the median Terrain Ruggedness Index (TRI). 'Plains' indicates TRI below the sample median; 'Mountains' indicates TRI above the median.

B. Spatial Variation of Key Variables

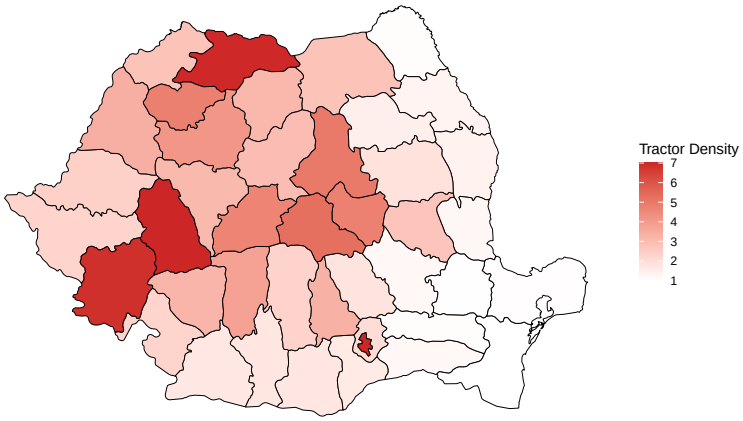


FIGURE A1: SPATIAL VARIATION OF TRACTOR DENSITY

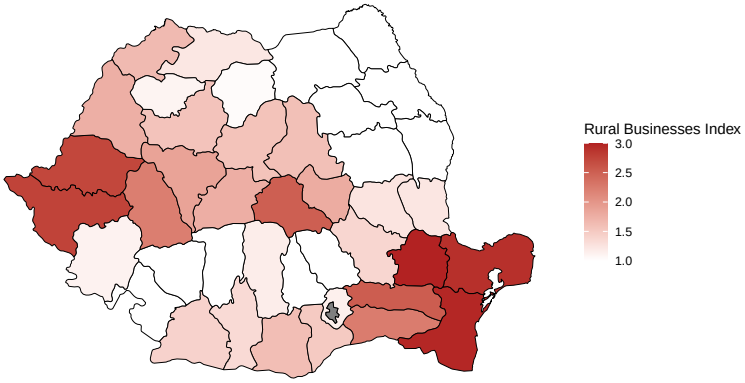


FIGURE A2: SPATIAL VARIATION OF RURAL BUSINESS FORMATION

Notes: Bucharest Municipality does not have rural population.

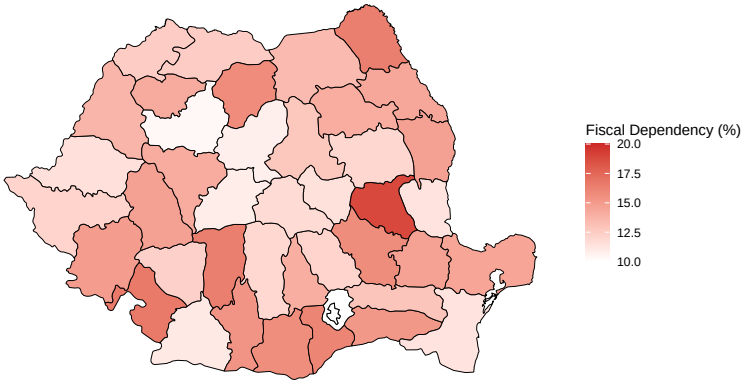


FIGURE A3: SPATIAL VARIATION OF FISCAL DEPENDENCY

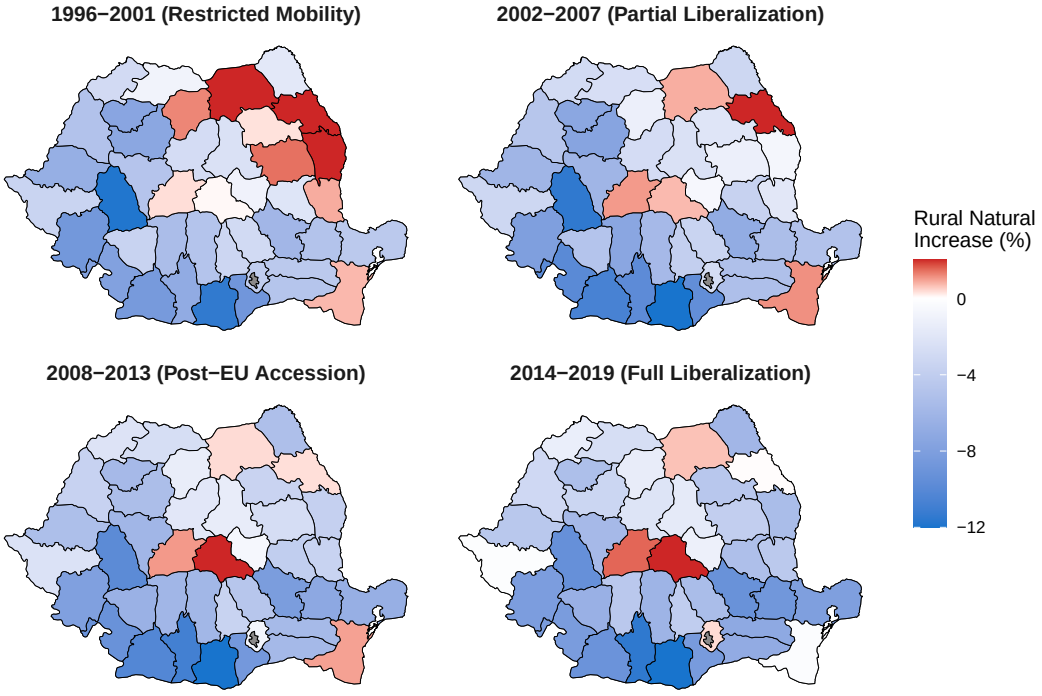


FIGURE A4: RURAL NATURAL INCREASE OVER TIME

Notes: Positive values entail more births than deaths and vice versa. Bucharest Municipality does not have rural population.

C. Robustness

TABLE C3: SYNTHETIC DIFFERENCE-IN-DIFFERENCES RESULTS: LOW-PSD FALSIFICATION TEST

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
SDiD Estimate ($\hat{\tau}^{SDiD}$)	-1.037***	0.399**	-0.060	0.005	-0.937
	(0.305)	(0.189)	(0.127)	(0.701)	(0.878)
Observations	648	621	648	648	648
Treated Counties	6	6	6	6	6
Donor Pool (Mountains)	21	21	21	21	21
Time Weights ($\hat{\lambda}_t$)	Estimated	Estimated	Estimated	Estimated	Estimated
Unit Weights ($\hat{\omega}_c$)	Estimated	Estimated	Estimated	Estimated	Estimated
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Standard Errors	Jackknife	Jackknife	Jackknife	Jackknife	Jackknife

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Jackknife standard errors are reported in parentheses. As a falsification test, this model matches the Low-PSD Plains sub-cohort to a Synthetic Mountain Baseline.

TABLE C4: ROBUSTNESS CHECK: EXTREME TRI SPLIT

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Extreme Plains × Post-2007	-2.368***	0.501***	-0.279	-0.439	-0.187
	(0.539)	(0.161)	(0.167)	(0.665)	(1.371)
Dependent Variable Mean	2.893	1.748	-0.267	-5.449	14.000
	(2.447)	(0.953)	(0.674)	(3.983)	(9.588)
Observations	460	460	460	459	460
Adjusted R^2	0.831	0.927	0.251	0.931	0.878
Adjusted Within R^2	0.299	0.251	0.051	0.113	0.078
Controls	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	20	20	20	20	20

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects, as well as control variables. This sample restricts the dataset to the top 25% most rugged (Mountains) and bottom 25% flattest (Plains) counties.

TABLE C5: ROBUSTNESS CHECK: EXTREME PSD PLAINS SPLIT

Dependent Variable:	**Tractor Density**		**Business Formation**		**External Migration**		**Rural Natural Increase**		**Fiscal Dependency**	
PSD Level:	**High** (1)	**Low** (2)	**High** (3)	**Low** (4)	**High** (5)	**Low** (6)	**High** (7)	**Low** (8)	**High** (9)	**Low** (10)
Plains × Post-2007	-1.563***	-1.500**	0.239	0.520***	1.567	-0.167	-2.322**	0.716	0.398	-0.927
	(0.305)	(0.545)	(0.242)	(0.162)	(1.590)	(0.173)	(0.945)	(0.826)	(0.755)	(0.981)
Observations	598	598	598	598	598	598	597	597	598	598
Adjusted R^2	0.804	0.773	0.905	0.925	0.366	0.317	0.923	0.890	0.882	0.879
Adjusted Within R^2	0.143	0.078	0.097	0.309	0.112	0.015	0.245	0.134	0.071	0.045
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	26	26	26	26	26	26	26	26	26	26

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects, as well as control variables. The model compares the extreme PSD Plains to the Mountains.

TABLE C6: ROBUSTNESS CHECK: TRUNCATED TIMELINE (2000–2019)

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Plains × Post-2007	-1.460***	0.273**	0.716	-0.945*	0.658
	(0.309)	(0.103)	(0.590)	(0.547)	(0.707)
Dependent Variable Mean	3.033	1.606	0.042	-4.515	11.000
	(2.164)	(0.866)	(1.968)	(3.704)	(6.214)
Observations	800	800	800	800	800
Adjusted R^2	0.831	0.939	0.364	0.915	0.793
Adjusted Within R^2	0.161	0.145	0.045	0.092	0.057
Controls	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects, as well as control variables. The sample timeline is truncated to 2000–2019 to test robustness against extreme years.

FIGURE B5: LEAVE-ONE-OUT ROBUSTNESS TEST FOR TRACTOR DENSITY

Notes: This figure presents point estimates and 95% confidence intervals for the baseline Difference-in-Differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

FIGURE B6: LEAVE-ONE-OUT ROBUSTNESS TEST FOR RURAL BUSINESS FORMATION

Notes: This figure presents point estimates and 95% confidence intervals for the baseline Difference-in-Differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

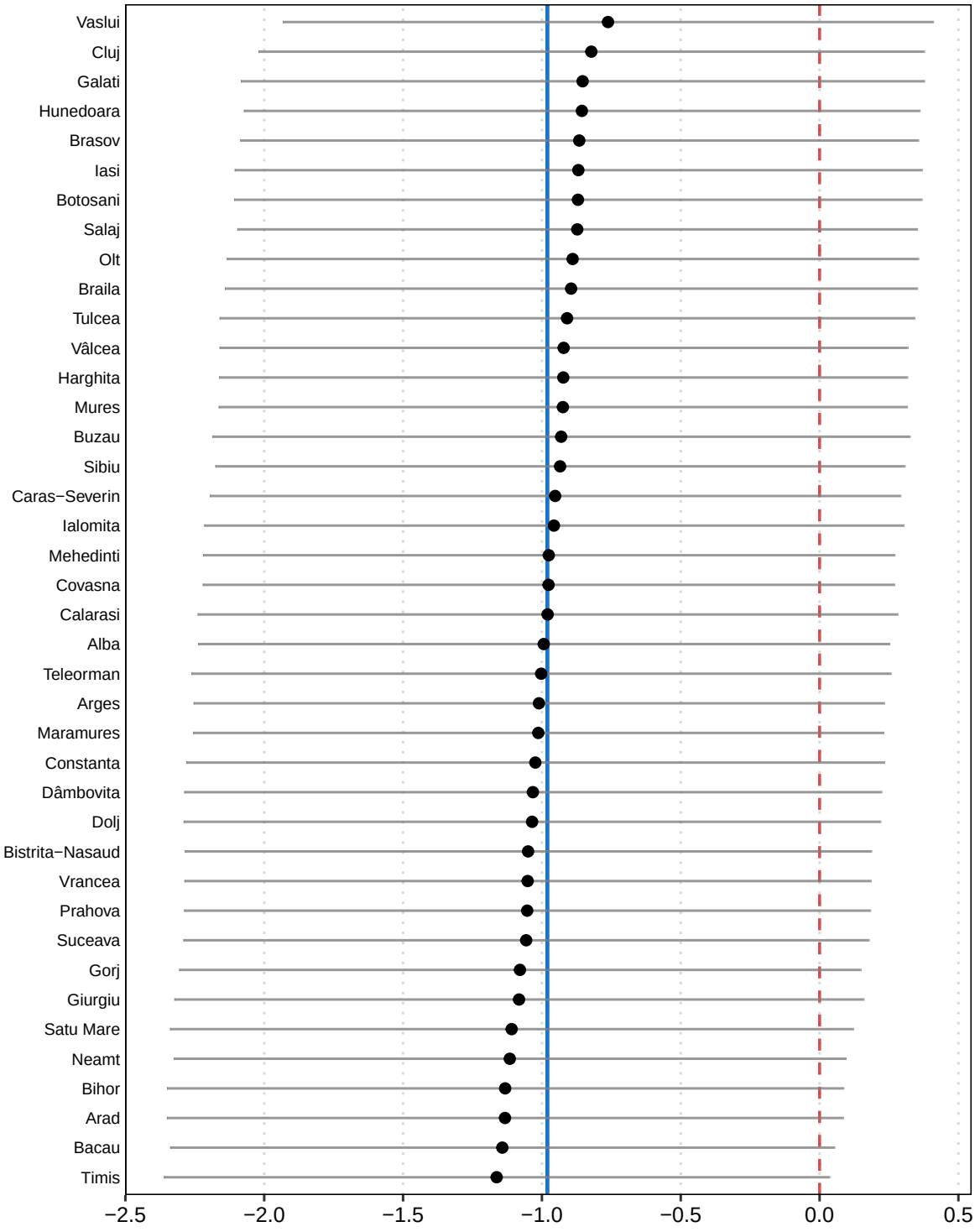


FIGURE B7: LEAVE-ONE-OUT ROBUSTNESS TEST FOR RURAL NATURAL INCREASE

Notes: This figure presents point estimates and 95% confidence intervals for the baseline Difference-in-Differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

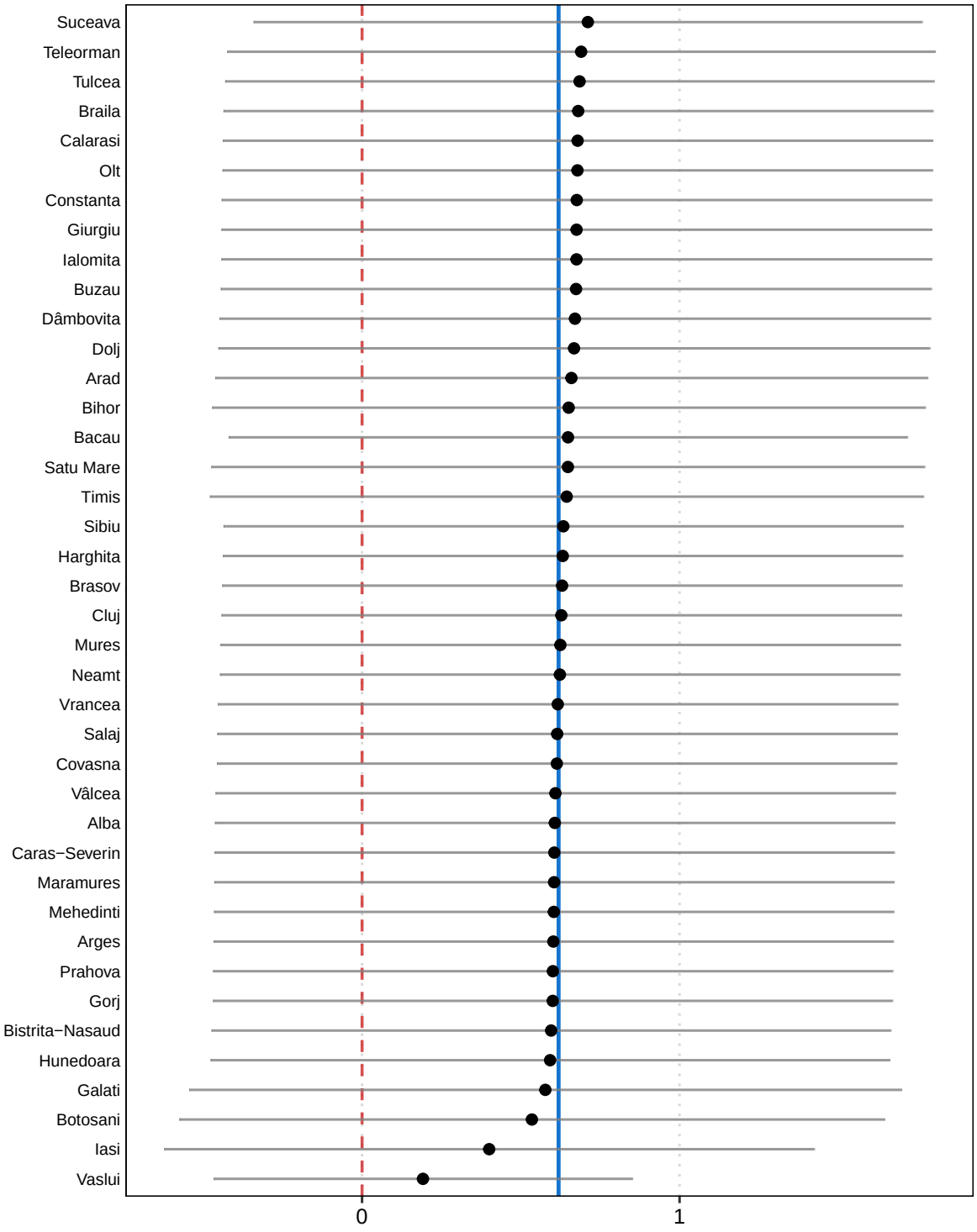


FIGURE B8: LEAVE-ONE-OUT ROBUSTNESS TEST FOR NET MIGRATION RATE

Notes: This figure presents point estimates and 95% confidence intervals for the baseline Difference-in-Differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

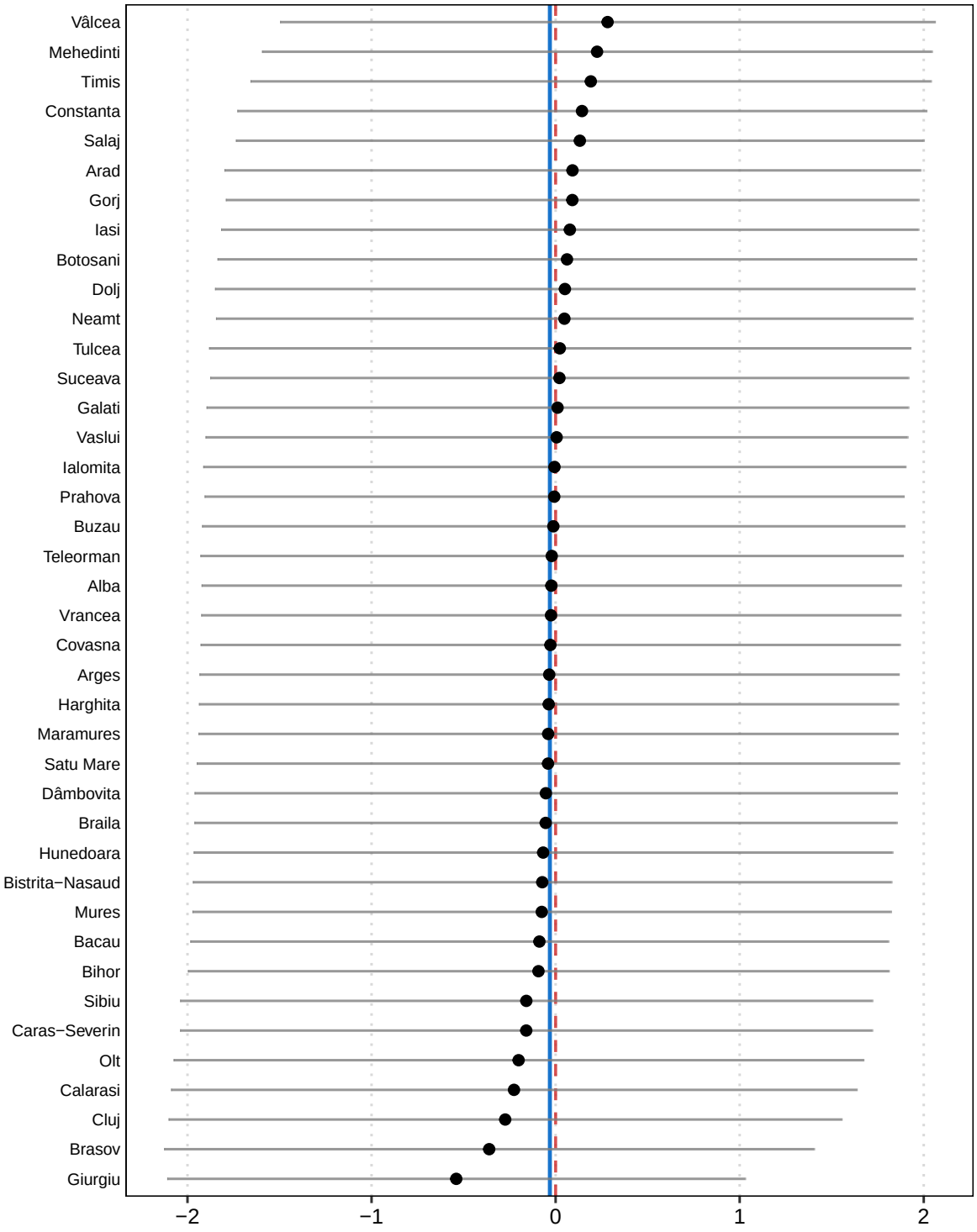


FIGURE B9: LEAVE-ONE-OUT ROBUSTNESS TEST FOR FISCAL DEPENDENCY

Notes: This figure presents point estimates and 95% confidence intervals for the baseline Difference-in-Differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

D. Identification Without Controls

TABLE D7: OLS REGRESSION RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	(1) **Privatization Rate**	(2) **Tractor Density**	(3) **Business Formation**	(4) **External Migration**	(5) **Rural Natural Increase**	(6) **Fiscal Dependency**	(7) **PSD Vote Share**
Plains Dummy	-4.745***	-1.790***	0.193	0.377**	-1.338	0.091	8.054**
	(1.356)	(0.432)	(0.211)	(0.190)	(1.221)	(0.568)	(3.455)
Privatization Rate		-0.006	-0.036**	0.015	0.053	-0.037	-0.088
		(0.028)	(0.014)	(0.009)	(0.050)	(0.035)	(0.161)
Dependent Variable Mean	94.399	2.589	1.386	-0.180	-4.301	12.892	32.125
	(7.378)	(1.631)	(0.744)	(1.078)	(3.777)	(10.115)	(13.075)
Observations	760	760	720	760	759	760	160
Adjusted R^2	0.404	0.378	0.244	0.083	0.059	0.903	0.276
Adjusted Within R^2	0.150	0.323	0.142	0.026	0.049	0.004	0.121
Controls	No	No	No	No	No	No	No
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County
No. of Counties	40	40	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include Time Fixed Effects.

TABLE D8: POOLED DIFFERENCE-IN-DIFFERENCES RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Plains × Post-2007	-1.613***	0.304**	0.619	-0.980	-0.031
	(0.334)	(0.119)	(0.527)	(0.602)	(0.916)
Dependent Variable Mean	2.842	1.546	-0.034	-4.407	13.638
	(2.059)	(0.851)	(1.818)	(3.756)	(9.541)
Observations	960	920	960	959	960
Adjusted R^2	0.815	0.915	0.314	0.891	0.888
Adjusted Within R^2	0.179	0.089	0.010	0.039	-0.001
Controls	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects.

TABLE D9: SPLIT-SAMPLE DIFFERENCE-IN-DIFFERENCES RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	**Tractor Density**		**Business Formation**		**External Migration**		**Rural Natural Increase**		**Fiscal Dependency**	
PSD Level:	**High** (1)	**Low** (2)	**High** (3)	**Low** (4)	**High** (5)	**Low** (6)	**High** (7)	**Low** (8)	**High** (9)	**Low** (10)
Plains × Post-2007	-1.672***	-1.484***	0.157	0.622***	0.966	-0.133	-1.794***	0.783	0.664	-1.537
	(0.335)	(0.416)	(0.131)	(0.147)	(0.742)	(0.161)	(0.590)	(0.771)	(1.047)	(0.998)
Observations	816	648	782	621	816	648	815	647	816	648
Adjusted R^2	0.816	0.778	0.905	0.927	0.322	0.340	0.912	0.876	0.887	0.898
Adjusted Within R^2	0.168	0.086	0.028	0.254	0.021	0.001	0.131	0.019	0.001	0.011
Controls	No	No	No	No	No	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	34	27	34	27	34	27	34	27	34	27

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects.

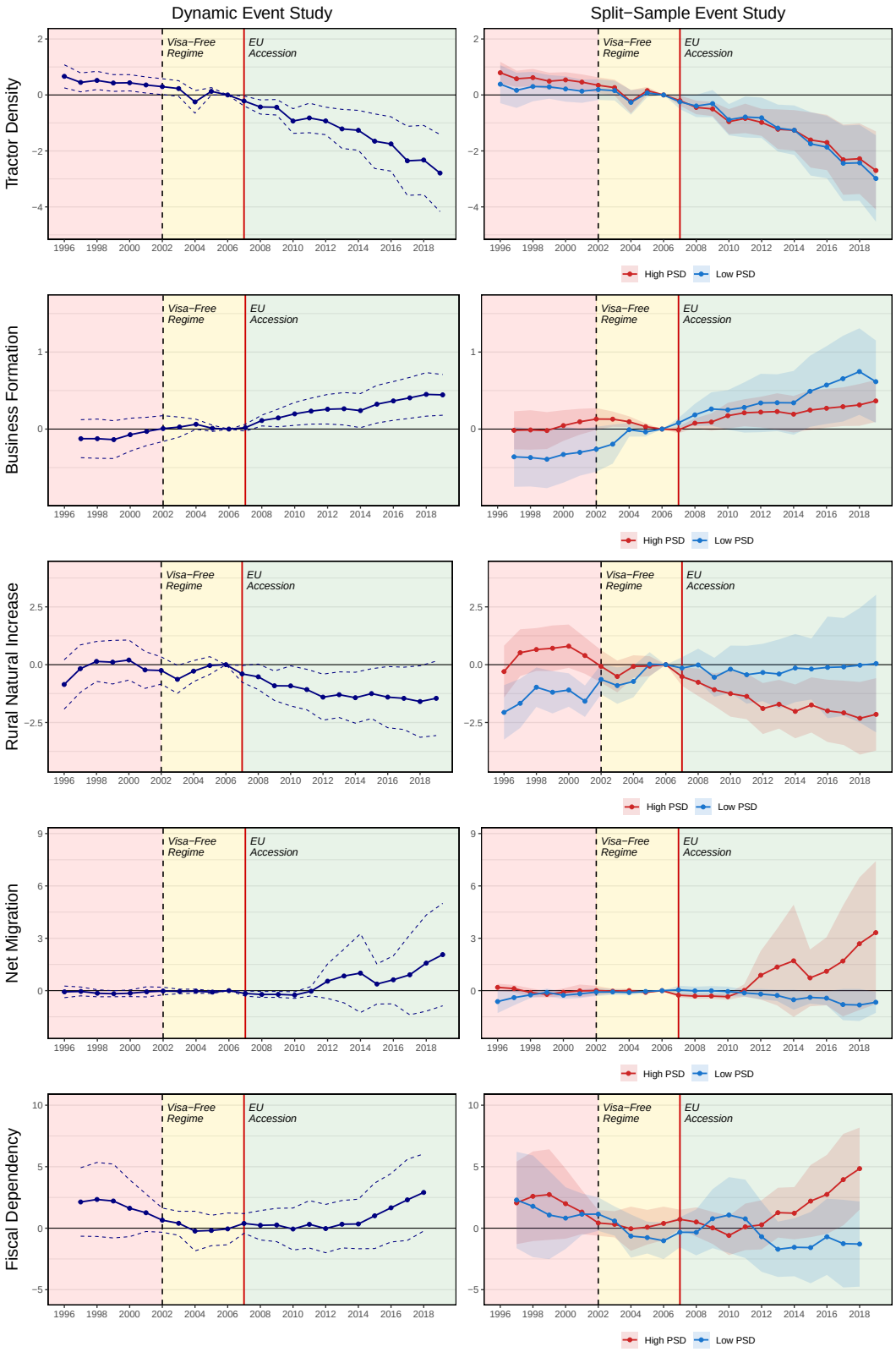


FIGURE C10: DYNAMIC & SPLIT-SAMPLE EVENT STUDIES: MODELS WITHOUT CONTROLS

Notes: Left column: aggregate event study estimates (Equation 4, but without controls). Right column: split-sample estimates partitioning the Plains into High- and Low-PSD cohorts. Solid lines and markers denote point estimates (β_k), with 95% confidence intervals shown as dashed lines (left) and shaded regions (right). To filter short-term volatility, Fiscal Dependency is smoothed via a 3-year centered moving average.

TABLE D10: SYNTHETIC DIFFERENCE-IN-DIFFERENCES RESULTS: LOW PSD PLAINS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
SDiD Estimate ($\hat{\tau}^{SDiD}$)	-1.037***	0.399**	-0.060	0.005	-0.937
	(0.305)	(0.189)	(0.127)	(0.701)	(0.878)
Observations	648	621	648	648	648
Treated Counties (Low PSD)	6	6	6	6	6
Donor Pool (Mountains)	21	21	21	21	21
Time Weights ($\hat{\lambda}_t$)	Estimated	Estimated	Estimated	Estimated	Estimated
Unit Weights ($\hat{\omega}_c$)	Estimated	Estimated	Estimated	Estimated	Estimated
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Standard Errors	Jackknife	Jackknife	Jackknife	Jackknife	Jackknife

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Jackknife standard errors are reported in parentheses. As a falsification test, this model matches Low-PSD Plains counties to a Synthetic Mountain Baseline.

TABLE D11: ROBUSTNESS CHECK: EXTREME TRI SPLIT WITHOUT CONTROLS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Extreme Plains × Post-2007	-2.463***	0.411**	-0.378	-0.428	-0.146
	(0.547)	(0.183)	(0.231)	(0.719)	(1.425)
Dependent Variable Mean	2.893	1.748	-0.267	-5.449	14.000
	(2.447)	(0.953)	(0.674)	(3.983)	(9.588)
Observations	480	460	480	479	480
Adjusted R^2	0.821	0.916	0.253	0.920	0.887
Adjusted Within R^2	0.279	0.131	0.028	0.010	0.000
Controls	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	20	20	20	20	20

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects.

TABLE D12: ROBUSTNESS CHECK: EXTREME PSD PLAINS SPLIT WITHOUT CONTROLS

Dependent Variable:	**Tractor Density**		**Business Formation**		**External Migration**		**Rural Natural Increase**		**Fiscal Dependency**	
PSD Level:	**High** (1)	**Low** (2)	**High** (3)	**Low** (4)	**High** (5)	**Low** (6)	**High** (7)	**Low** (8)	**High** (9)	**Low** (10)
Plains × Post-2007	-1.749***	-1.363***	0.251	0.624***	1.266	-0.077	-2.503***	0.980	0.425	-1.210
	(0.347)	(0.432)	(0.235)	(0.173)	(1.642)	(0.167)	(0.763)	(0.861)	(0.899)	(1.057)
Observations	624	624	598	598	624	624	623	623	624	624
Adjusted R^2	0.794	0.769	0.901	0.917	0.306	0.343	0.912	0.871	0.894	0.896
Adjusted Within R^2	0.107	0.067	0.049	0.233	0.027	0.001	0.161	0.028	0.001	0.007
Controls	No	No	No	No	No	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	26	26	26	26	26	26	26	26	26	26

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects.

TABLE D13: ROBUSTNESS CHECK: TRUNCATED TIMELINE (2000–2019) WITHOUT CONTROLS

Dependent Variable:	(1) **Tractor Density**	(2) **Business Formation**	(3) **External Migration**	(4) **Rural Natural Increase**	(5) **Fiscal Dependency**
Plains × Post-2007	-1.488***	0.266**	0.600	-0.987*	0.578
	(0.313)	(0.107)	(0.528)	(0.533)	(0.769)
Dependent Variable Mean	3.033	1.606	0.042	-4.515	11.000
	(2.164)	(0.866)	(1.968)	(3.704)	(6.214)
Observations	800	800	800	800	800
Adjusted R^2	0.828	0.935	0.343	0.911	0.782
Adjusted Within R^2	0.144	0.081	0.009	0.047	0.002
Controls	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Standard errors are clustered at the county level. All models include County and Time Fixed Effects.